

Detailed Analytical Analysis

Detailed analysis of the business questions explored during the project, including the initial analytics, deeper driver investigations, statistical validation, and conclusions drawn from the evidence.

1. Initial Analytics

1.1 Operational Performance & Cancellations

1.1.1 Is Kitchen Preparation Time associated with customer cancellations?

Why This Matters :

Customer cancellations can lead to lost revenue, reduced operational efficiency, and a poor customer experience. If orders with longer Kitchen Preparation Times are more likely to be cancelled, it could indicate that preparation delays contribute to customer dissatisfaction. Understanding whether such an association exists helps evaluate whether kitchen operations are an important factor in customer cancellation behaviour.

EDA :

Customer Cancelled	Number of Orders	Mean KPT (Minutes)	Median KPT (Minutes)	Standard Deviation
No	21,004	17.33	16.33	6.28
Yes	22	16.29	14.92	6.35

EDA Interpretation :

Customer-cancelled orders had a slightly lower median KPT (**14.92 minutes**) than non-cancelled orders (**16.33 minutes**). The mean showed a similar pattern, at **16.29 minutes versus 17.33 minutes**, while the variability was similar between the two groups.

The difference was small, so the descriptive results do not suggest an obvious relationship between KPT and customer cancellation. However, EDA alone cannot determine whether the observed difference is statistically meaningful. Therefore, a statistical test was performed.

Assumption / Test Selection :

- KPT is a continuous variable.
- Customer cancellation status forms two independent groups.
- KPT was positively skewed.
- Therefore, a **Mann-Whitney U test** was used to compare the two groups.

Statistical Hypothesis :

H₀: There is no significant difference in Kitchen Preparation Time between customer-cancelled and non-cancelled orders.

H₁: There is a significant difference in Kitchen Preparation Time between customer-cancelled and non-cancelled orders.

Statistical Validation :

Metric	Result
U Statistic	202,185.50

Metric	Result
p-value	0.3105
Significance Level (α)	0.05

Since the p-value (**0.3105**) is greater than **0.05**, we fail to reject the null hypothesis. There is therefore **insufficient statistical evidence to conclude that KPT differs between customer-cancelled and non-cancelled orders**.

Effect Size :

The rank-biserial correlation was **-0.125**, indicating a **small effect**. This suggests that the difference between the two groups is also weak in practical terms.

Investigation Conclusion :

The analysis does not provide evidence that **Kitchen Preparation Time is meaningfully associated with customer cancellations** in this dataset. Although customer-cancelled orders had slightly lower KPT descriptively, the difference was neither statistically significant nor practically strong.

Therefore, KPT does not appear to be a meaningful standalone driver of customer cancellation in the observed data.

1.1.2 Is customer cancellation associated with the time of day in which an order is placed?

Why This Matters :

Operational conditions such as restaurant workload, rider availability, and order volume can vary across different times of the day. Understanding whether customer cancellations are associated with time of day can help identify whether certain periods may require closer operational attention.

EDA :

Time of Day	Total Orders	Customer-Cancelled Orders	Customer Cancellation Rate (%)
Evening / Dinner	12,463	39	0.31
Late Night	2,672	18	0.67
Lunch / Afternoon	6,186	32	0.52

EDA Interpretation :

Customer cancellation rates varied across the three time periods. **Late Night had the highest cancellation rate (0.67%)**, followed by Lunch/Afternoon (0.52%), while Evening/Dinner had the lowest rate (0.31%).

Although Evening/Dinner had the largest number of customer-cancelled orders, it also had the highest order volume. Looking at cancellation **rates** therefore gives a different picture from looking at cancellation counts alone.

These descriptive differences suggest that cancellation behaviour may vary by time of day, so statistical testing was used to determine whether the observed differences were statistically meaningful.

Statistical Hypotheses :

H₀: There is no association between time of day and customer cancellations.

H₁: There is an association between time of day and customer cancellations.

Assumption / Test Selection :

The **Chi-Square Test of Independence** was used because both variables—time of day and customer cancellation status—are categorical. The observations were treated as independent, and all expected cell frequencies were greater than 5.

Statistical Validation :

The Chi-Square Test of Independence showed a statistically significant association between time of day and customer cancellations.

Metric	Result
χ^2 Statistic	8.98
p-value	0.0112
Significance Level (α)	0.05

Since $p = 0.0112 < 0.05$, the null hypothesis was rejected. This provides statistical evidence that customer cancellation rates differed across the time-of-day categories.

Effect Size :

Cramér's V = 0.0205, indicating a **negligible association** between time of day and customer cancellation.

Therefore, although the difference is statistically detectable, the practical strength of the relationship is very weak.

Investigation Conclusion :

Customer cancellation rates were statistically associated with time of day, with **Late Night showing the highest observed cancellation rate**. However, the negligible effect size indicates that time of day has **very limited practical influence** on customer cancellation behaviour in this dataset.

Therefore, time of day should not be treated as a major standalone driver of customer cancellation.

1.1.3 Do orders cancelled by Zomato exhibit different Kitchen Preparation Times compared with orders that were not cancelled by Zomato?

Why This Matters :

Longer Kitchen Preparation Times may delay order fulfilment and could potentially be associated with platform-initiated cancellations. Understanding whether KPT differs between Zomato-cancelled and non-cancelled orders helps determine whether kitchen preparation is a meaningful operational factor related to Zomato cancellations.

EDA :

Zomato Cancelled	Orders	Mean KPT (Minutes)	Median KPT (Minutes)	Standard Deviation
No	20,960	17.34	16.35	6.28
Yes	66	15.65	14.72	7.81

Orders cancelled by Zomato had a slightly lower Kitchen Preparation Time than orders that were not cancelled. The median KPT was **14.72 minutes** for Zomato-cancelled orders compared with **16.35 minutes** for non-cancelled orders. The mean showed a similar pattern, at **15.65 minutes versus 17.34 minutes**.

Although a difference is visible in the descriptive statistics, EDA alone cannot determine whether this difference is statistically meaningful. Therefore, a statistical test was performed.

Assumption / Test Selection :

- KPT is a continuous numerical variable.
- Zomato cancellation status forms two independent groups.
- KPT was not assumed to follow a normal distribution.
- Therefore, a **Mann-Whitney U test** was used to compare the two groups.

Statistical Hypotheses :

H₀: The distribution of KPT is the same for Zomato-cancelled and non-cancelled orders.

H₁: The distribution of KPT differs between Zomato-cancelled and non-cancelled orders.

Statistical Validation :

Test	Test Statistic	p-value	Decision ($\alpha = 0.05$)
Mann-Whitney U Test	563,070	0.0090	Reject H ₀

Since **p = 0.009 < 0.05**, the null hypothesis was rejected. There is statistical evidence that the KPT distributions differ between Zomato-cancelled and non-cancelled orders.

Effect Size :

Rank-Biserial Correlation = -0.186

The effect size indicates a **small practical difference** between the two groups. Therefore, although the difference is statistically significant, its practical magnitude is limited.

Investigation Conclusion :

Zomato-cancelled orders had statistically different KPT from non-cancelled orders, but the difference was **small in practical terms**. Interestingly, Zomato-cancelled orders actually had a **lower median KPT (14.72 minutes)** than non-cancelled orders (**16.35 minutes**).

Therefore, **KPT alone does not appear to be a strong operational indicator of Zomato cancellation behaviour** during the study period.

1.1.4 Do orders cancelled by Zomato exhibit different rider wait times compared with orders that were not cancelled by Zomato?

Why This Matters :

Rider Wait Time represents the time a delivery partner waits at the restaurant before collecting an order. Longer waiting times may indicate operational inefficiencies that could be associated with order cancellations. Understanding whether Rider Wait Time differs between Zomato-cancelled and non-cancelled orders helps determine whether rider waiting is an important operational factor related to Zomato cancellations.

EDA :

Zomato Cancelled	Orders	Mean Rider Wait (Minutes)	Median Rider Wait (Minutes)	Standard Deviation
No	21,097	4.83	3.10	4.98
Yes	56	4.81	3.50	5.08

EDA Interpretation :

Rider wait times were very similar between Zomato-cancelled and non-cancelled orders. The median rider wait was **3.50 minutes** for Zomato-cancelled orders compared with **3.10 minutes** for non-cancelled orders. The mean values were also almost identical at **4.81 minutes and 4.83 minutes**, respectively.

The descriptive results therefore suggest very little difference between the two groups. A statistical test was used to determine whether the observed difference was statistically meaningful.

Assumption / Test Selection :

- Rider Wait Time is a continuous numerical variable.
- Zomato cancellation status forms two independent groups.
- Rider Wait Time was not assumed to follow a normal distribution.
- Therefore, a **Mann-Whitney U test** was used to compare the two groups.

Statistical Hypotheses :

H₀: The distribution of rider wait time is the same for Zomato-cancelled and non-cancelled orders.

H₁: The distribution of rider wait time differs between Zomato-cancelled and non-cancelled orders.

Statistical Validation :

Metric	Result
Test Used	Mann-Whitney U Test
U Statistic	581,343
p-value	0.8373
Significance Level (α)	0.05
Decision	Fail to reject H ₀

Since **p = 0.8373 > 0.05**, the null hypothesis was not rejected. There is insufficient statistical evidence to conclude that rider wait times differ between Zomato-cancelled and non-cancelled orders.

Effect Size :

Rank-Biserial Correlation = -0.0159

The effect size is **negligible**, indicating virtually no practical difference in rider wait times between the two groups.

Investigation Conclusion :

The analysis found **no statistically significant difference** in rider wait times between Zomato-cancelled and non-cancelled orders (**U = 581,343, p = 0.8373**). The negligible effect size (**-0.0159**) further indicates that the two groups were practically very similar.

Therefore, **rider wait time does not appear to be associated with Zomato cancellation behaviour within the study period.**

1.1.5 Is there an association between Time of Day and orders being cancelled by Zomato?

Why This Matters :

The time of day can influence food delivery operations because customer demand and operating conditions may vary throughout the day. Factors such as rider availability, restaurant workload, and traffic conditions can differ across periods. Understanding whether Time of Day is associated with Zomato cancellations helps determine whether cancellation patterns vary across different time periods.

EDA :

Time of Day	Total Orders	Zomato Cancelled Orders	Cancellation Rate (%)
Late Night	2,672	24	0.898

Time of Day	Total Orders	Zomato Cancelled Orders	Cancellation Rate (%)
Evening / Dinner	12,463	40	0.321
Lunch / Afternoon	6,186	17	0.275

EDA Interpretation :

Zomato cancellation rates varied across the three time periods. **Late Night had the highest cancellation rate (0.898%)**, followed by Evening/Dinner (0.321%) and Lunch/Afternoon (0.275%).

Although these differences are visible descriptively, EDA alone cannot determine whether they represent a genuine association or random variation. Therefore, a statistical test was conducted.

Statistical Hypotheses :

H₀: There is no association between Time of Day and orders being cancelled by Zomato.

H₁: There is an association between Time of Day and orders being cancelled by Zomato.

Assumption / Test Selection :

- Time of Day and Zomato cancellation status are categorical variables.
- The observations are independent.
- The expected cell frequencies satisfy the assumptions of the Chi-Square Test of Independence.

Therefore, a **Chi-Square Test of Independence** was used.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Test of Independence
χ^2 Statistic	21.915
Degrees of Freedom	2
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H ₀

Since **p < 0.001**, the null hypothesis was rejected. There is statistical evidence of an association between **Time of Day and Zomato cancellations**.

Effect Size :

Cramér's V = 0.0321

The effect size indicates a **negligible association** between Time of Day and Zomato cancellations. Therefore, although the relationship is statistically significant, its practical strength is very weak.

Investigation Conclusion :

Zomato cancellation rates differed statistically across time periods, with **Late Night showing the highest observed cancellation rate**. However, the negligible effect size indicates that Time of Day has **very limited practical influence** on Zomato cancellation behaviour.

Therefore, **Time of Day should not be treated as a meaningful standalone driver of Zomato cancellations during the study period**.

1.1.6 Is there an association between the restaurant subzone and orders being cancelled by Zomato?

Why This Matters :

Restaurant subzones may differ in operational conditions such as restaurant density, rider availability, traffic, and local demand. Understanding whether Zomato cancellation rates vary across subzones helps determine whether location at the restaurant level is associated with platform cancellations.

EDA :

Restaurant Subzone	Total Orders	Zomato-Cancelled Orders	Cancellation Rate (%)
Sector 4	6,530	34	0.521
Greater Kailash 2 (GK2)	7,380	28	0.379
Sector 135	2,442	9	0.369
DLF Phase 1	3,686	9	0.244
Vasant Kunj	920	1	0.109
Shahdara	360	0	0.000
Chittaranjan Park	2	0	0.000
Sikandarpur	1	0	0.000

EDA Interpretation :

Zomato cancellation rates varied across restaurant subzones. **Sector 4 recorded the highest cancellation rate (0.521%)**, followed by Greater Kailash 2 (0.379%) and Sector 135 (0.369%). Vasant Kunj recorded a lower rate of 0.109%, while Shahdara, Chittaranjan Park, and Sikandarpur recorded no Zomato cancellations.

However, the latter subzones had **very small order volumes**, so their observed rates should not be overinterpreted.

Since descriptive differences alone cannot determine whether the observed variation is statistically meaningful, a Chi-Square Test of Independence was conducted.

Statistical Hypotheses :

H₀: There is no association between restaurant subzone and orders being cancelled by Zomato.

H₁: There is an association between restaurant subzone and orders being cancelled by Zomato.

Assumption / Test Selection :

- Restaurant Subzone and Zomato Cancellation Status are categorical variables.
- Observations are independent.
- No expected frequency was below 1, and more than 80% of expected frequencies were at least 5.

Therefore, a **Chi-Square Test of Independence** was used.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Test of Independence
χ^2 Statistic	10.978
Degrees of Freedom	5
p-value	0.0518

Metric	Result
Significance Level (α)	0.05
Decision	Fail to reject H_0

Since $p = 0.0518 > 0.05$, the null hypothesis was not rejected. There is insufficient statistical evidence to conclude that restaurant subzone is associated with Zomato cancellations.

Effect Size :

Cramér's V = 0.0256

This indicates a **negligible association**, suggesting that restaurant subzone has very little practical relationship with Zomato cancellation behaviour.

Investigation Conclusion :

The analysis did not find a statistically significant association between **restaurant subzone and Zomato cancellations**. The negligible effect size also indicates that the practical relationship is very weak.

Therefore, **restaurant subzone alone does not appear to be an important factor associated with Zomato cancellations during the study period**. The higher observed cancellation rates in some subzones are not strong enough, based on this analysis, to justify treating location as a standalone driver.

1.1.7 Could incorrect or missed order-ready markings be contributing to longer rider wait times?

Why This Matters :

Accurate order-ready marking helps restaurants and delivery riders coordinate the pickup process smoothly. If an order is marked incorrectly or the ready status is missed, riders may arrive before the order is actually ready, potentially increasing their waiting time. Understanding whether rider wait time differs across order-ready marking statuses can therefore help identify inefficiencies in the restaurant pickup process.

EDA :

Order Ready Marked Status	Number of Orders	Mean Rider Wait Time (Minutes)	Median Rider Wait Time (Minutes)	Standard Deviation
Correctly	19,054	4.44	2.50	4.73
Incorrectly	1,890	8.63	7.20	5.63
Missed	209	5.75	3.70	6.24

EDA Interpretation :

Rider wait time differed noticeably across the order-ready marking statuses. Orders marked **Incorrectly** had the highest mean and median rider wait times (**8.63 and 7.20 minutes**), while orders marked **Correctly** had the lowest (**4.44 and 2.50 minutes**). Orders with **Missed** markings showed intermediate wait times, although this group contained substantially fewer observations.

The descriptive results therefore suggest that incorrect or missed order-ready markings may be linked to longer rider waiting. Statistical testing was used to determine whether these differences were statistically meaningful.

Assumption / Test Selection :

Rider wait time was not normally distributed within the three order-ready marking groups, based on the Shapiro-Wilk tests, with all groups producing **p-values < 0.05**.

Therefore, the normality assumption required for a One-Way ANOVA was not satisfied. A **Kruskal-Wallis H test**, the non-parametric alternative for comparing more than two independent groups, was therefore used.

Statistical Hypotheses :

H₀: The distribution of rider wait time is the same across the different order-ready marking statuses.

H₁: Rider wait time differs across at least one of the order-ready marking statuses.

Statistical Validation :

The Kruskal-Wallis H test showed a statistically significant difference in rider wait time across the order-ready marking groups.

Metric	Result
Test Used	Kruskal-Wallis H Test
H Statistic	1465.7971
p-value	< 0.001
Decision	Reject H ₀

Since **p < 0.001**, the null hypothesis was rejected. This provides statistical evidence that rider wait time differs across the order-ready marking statuses.

Effect Size :

Epsilon Squared (ϵ^2) = 0.0692

The effect size indicates a **moderate practical difference** in rider wait time across the order-ready marking statuses.

Investigation Conclusion :

Rider wait time differed significantly across order-ready marking statuses. Orders marked **Incorrectly** had substantially higher rider wait times than correctly marked orders, with median wait times of **7.20 minutes versus 2.50 minutes**.

The statistical test confirmed that these differences were unlikely to be due to random variation, while the **moderate effect size** indicates that the relationship is meaningful in practical terms.

This makes order-ready marking one of the more important operational patterns identified in the analysis and suggests that improving the accuracy of ready-status marking could be worth investigating as a way to reduce rider waiting.

1.1.8 Do longer-distance orders show higher customer cancellation rates?

Why This Matters :

Longer delivery distances may increase the friction associated with an order, such as longer expected delivery times or greater delivery uncertainty. This could potentially make customers more likely to cancel.

Understanding whether cancellation rates vary with delivery distance helps determine whether distance-related delivery conditions should be considered when investigating customer cancellation behaviour.

EDA :

Distance Group	Total Orders	Customer Cancelled	Customer Cancellation Rate (%)
Nearby (≤ 2 km)	7,553	25	0.33
Short (3–5 km)	7,762	35	0.45

Distance Group	Total Orders	Customer Cancelled	Customer Cancellation Rate (%)
Medium (6–8 km)	4,103	13	0.32
Long (≥9 km)	1,903	16	0.84

EDA Interpretation :

Customer cancellation rates were relatively low across all distance groups. However, **Long-distance orders (≥9 km)** recorded the highest cancellation rate at **0.84%**, compared with **0.45%** for Short-distance, **0.33%** for Nearby, and **0.32%** for Medium-distance orders.

This suggests that longer-distance orders may experience relatively higher customer cancellation rates. However, the descriptive differences alone cannot determine whether the observed variation is statistically meaningful.

Statistical Hypotheses :

H₀: Delivery distance group and customer cancellation status are independent.

H₁: Delivery distance group and customer cancellation status are associated.

Assumption / Test Selection :

The **Chi-Square Test of Independence** was used because both variables are categorical:

- Distance Group
- Customer Cancellation Status

The minimum expected cell frequency was **7.94**, and no expected frequency was below 5, satisfying the expected-frequency requirement for the test.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Test of Independence
χ^2 Statistic	10.7704
Degrees of Freedom	3
p-value	0.0130
Significance Level (α)	0.05
Decision	Reject H ₀

Since **p = 0.0130 < 0.05**, the null hypothesis was rejected. There is statistical evidence of an association between **delivery distance group and customer cancellation status**.

Effect Size :

Cramér's V = 0.0225

This indicates a **negligible association** between delivery distance and customer cancellation.

Therefore, although the relationship is statistically detectable, delivery distance explains very little of the variation in customer cancellation behaviour.

Additional Investigation :

Residual analysis showed that the **Long-distance group had more customer cancellations than expected under independence**, with a standardized residual of **2.86**. This indicates that the Long-distance category contributed most strongly to the observed association.

Investigation Conclusion :

Long-distance orders (≥ 9 km) had the highest observed customer cancellation rate at **0.84%**, and statistical testing found an association between distance group and customer cancellation ($\chi^2 = 10.7704$, $p = 0.0130$).

However, the overall effect size was **negligible (Cramér's V = 0.0225)**. Therefore, while long-distance orders are worth monitoring because their cancellation rate is relatively higher, **delivery distance does not appear to be a meaningful standalone driver of customer cancellations during the study period.**

1.1.9 Does kitchen preparation time vary across different times of the day?

Why This Matters :

Kitchen Preparation Time is an important operational metric because it affects order fulfilment and delivery efficiency. Customer demand can vary throughout the day, which may also affect restaurant workload and preparation efficiency. Understanding whether KPT changes across different periods can help identify whether restaurants need to plan resources differently during specific times.

EDA :

Time of Day	Number of Orders	Mean KPT (Minutes)	Median KPT (Minutes)	Standard Deviation
Evening / Dinner	12,307	17.42	16.33	6.30
Late Night	2,627	16.45	15.32	7.17
Lunch / Afternoon	6,092	17.53	16.82	5.79

EDA Interpretation :

Kitchen Preparation Time showed some variation across the different time periods. **Lunch/Afternoon** had the highest mean and median KPT (**17.53 and 16.82 minutes**), followed by Evening/Dinner (**17.42 and 16.33 minutes**). **Late Night** had the lowest values (**16.45 and 15.32 minutes**).

However, the differences were relatively small. Statistical testing was therefore used to determine whether the observed variation was statistically meaningful.

Statistical Hypotheses :

H₀: The distribution of Kitchen Preparation Time is the same across the different time-of-day categories.

H₁: Kitchen Preparation Time differs across at least one of the time-of-day categories.

Assumption / Test Selection :

A **Kruskal–Wallis H test** was used to compare Kitchen Preparation Time across the three independent time-of-day groups. This non-parametric test was appropriate for comparing the distributions of a continuous variable across more than two groups without relying on a normality assumption.

Statistical Validation :

Metric	Result
Test Used	Kruskal–Wallis H Test
H Statistic	151.57
p-value	< 0.001
Decision	Reject H ₀

Since $p < 0.001$, the null hypothesis was rejected. There is statistical evidence that Kitchen Preparation Time differs across at least one of the time-of-day categories.

Effect Size :

Epsilon Squared (ϵ^2) = 0.0071

The effect size indicates a **negligible practical association** between time of day and Kitchen Preparation Time. Therefore, although the difference is statistically detectable, its practical magnitude is very small.

Investigation Conclusion :

Kitchen Preparation Time differed statistically across time-of-day categories, with **Lunch/Afternoon showing the highest typical preparation time and Late Night the lowest**. However, the negligible effect size indicates that these differences have **minimal practical importance**.

Therefore, **time of day does not appear to be a meaningful standalone driver of Kitchen Preparation Time**, and preparation times remained broadly consistent across the different periods during the study period.

1.1.10 Does rider wait time vary across customer rating levels ?

Why This Matters :

Customer ratings provide an indication of customer satisfaction and overall service quality. Rider wait time represents the time spent during the restaurant pickup stage and could potentially affect the delivery experience. Examining whether rider wait time differs across rating levels helps determine whether delays during pickup are reflected in customer feedback.

EDA :

Customer Rating	Number of Rated Orders	Mean Rider Wait Time (Minutes)	Median Rider Wait Time (Minutes)	Standard Deviation
1	176	4.46	3.10	4.64
2	82	4.68	2.05	5.36
3	144	4.74	3.10	4.62
4	356	4.53	2.90	4.66
5	1,724	4.93	3.00	5.22

EDA Interpretation :

Rider wait time showed only minor variation across customer rating levels. Mean rider wait time ranged from **4.46 to 4.93 minutes**, while median wait time remained close to **3 minutes** for most rating groups.

There was also **no clear increasing or decreasing pattern** in rider wait time across the rating levels. The descriptive results therefore do not suggest a strong relationship between rider waiting and customer ratings. Statistical testing was used to determine whether the observed differences were statistically meaningful.

Assumption / Test Selection :

Rider wait time was not normally distributed across the rating groups based on the normality assessment, with $p < 0.05$ in one or more groups.

Because the normality assumption for One-Way ANOVA was not satisfied, a **Kruskal-Wallis H test** was used to compare rider wait time across the five independent customer-rating groups.

Statistical Hypotheses :

H₀: The central tendency of rider wait time is the same across all customer rating levels.

H₁: At least one customer rating level has a different central tendency of rider wait time.

Statistical Validation :

Metric	Result
Test Used	Kruskal–Wallis H Test
H Statistic	2.9576
p-value	0.5649
Significance Level (α)	0.05
Decision	Fail to reject H ₀

Since **p = 0.5649 > 0.05**, the null hypothesis was not rejected. There is insufficient statistical evidence to conclude that rider wait time differs across customer rating levels.

Effect Size :

Epsilon Squared (ϵ^2) = 0.000

The effect size indicates a **negligible practical association** between rider wait time and customer ratings.

Investigation Conclusion :

The analysis found **no statistically significant difference in rider wait time across customer rating levels**, and the effect size was effectively zero.

Therefore, **rider wait time alone does not appear to be a meaningful factor associated with customer rating behaviour in this dataset**. The observed rating differences are unlikely to be explained by variation in rider waiting time during the restaurant pickup stage.

1.1.11 Do Zomato-cancelled orders differ in order value from non-cancelled orders?

Why This Matters :

Order value is directly related to transaction value and overall platform revenue. If cancelled orders have systematically different order values from non-cancelled orders, it could indicate that order value is associated with Zomato's cancellation behaviour. Understanding this helps determine whether order value should be considered when investigating platform cancellations.

EDA :

Zomato Cancelled	Number of Orders	Mean Order Value	Median Order Value	Minimum Order Value	Maximum Order Value	Standard Deviation
No	21,240	₹682.38	₹597.45	₹52.50	₹12,663.00	₹463.35
Yes	81	₹744.31	₹544.95	₹145.95	₹6,025.27	₹836.47

EDA Interpretation :

Total Order Amount was highly right-skewed, so the **median** was used as the primary measure of central tendency. The median order value was **₹597.45** for orders not cancelled by Zomato, compared with **₹544.95** for Zomato-cancelled orders.

Although the median values differed somewhat, the descriptive analysis alone cannot determine whether this difference is statistically meaningful. Therefore, a statistical test was used to compare the two groups.

Statistical Hypotheses :

H₀: There is no significant difference in total order amount between orders cancelled by Zomato and orders that were not cancelled.

H₁: There is a significant difference in total order amount between orders cancelled by Zomato and orders that were not cancelled.

Assumption / Test Selection :

- Total Order Amount is a continuous numerical variable.
- Zomato-cancelled and non-cancelled orders form two independent groups.
- Total Order Amount is highly right-skewed and does not require a normality assumption for the selected test.

Therefore, a **Mann-Whitney U test** was used to compare the two groups.

Statistical Validation :

Metric	Result
Test Used	Mann-Whitney U Test
U Statistic	791,036
p-value	0.2108
Significance Level (α)	0.05
Decision	Fail to reject H ₀

Since **p = 0.2108 > 0.05**, the null hypothesis was not rejected. There is insufficient statistical evidence to conclude that order value differs between Zomato-cancelled and non-cancelled orders.

Effect Size :

Rank-Biserial Correlation = -0.08

The effect size indicates a **negligible practical difference** in order value between the two groups.

Investigation Conclusion :

The analysis found **no statistically significant difference in order value** between Zomato-cancelled and non-cancelled orders (**U = 791,036, p = 0.2108**). The negligible effect size (**-0.08**) also indicates that the practical difference between the groups is very small.

Therefore, **order value does not appear to be an important standalone factor associated with Zomato cancellations during the study period.**

1.1.12 How much gross order value is associated with cancelled orders, and what restaurant compensation is recorded in connection with these cancellations ?

Why This Matters :

Cancelled orders can create business exposure through unrealized order value and potential compensation paid to restaurant partners. Quantifying the gross order value associated with cancellations and the compensation recorded against them helps assess the scale of this exposure and whether it differs between customer- and Zomato-initiated cancellations.

Analytical Approach :

Cancelled orders were identified using the two cancellation flags:

- **Customer Cancelled = "Yes"**

- `Zomato_Cancelled = "Yes"`

The two cancellation flags were mutually exclusive in the dataset, resulting in:

- **89 customer-cancelled orders**
- **81 Zomato-cancelled orders**
- **170 total cancelled orders**

`Total Order Amount` was used to measure the **gross order value associated with cancelled orders**, while `Restaurant compensation (Cancellation)` was used to measure **recorded restaurant compensation**.

Missing compensation values were retained as missing rather than automatically treated as zero.

Important: Gross order value associated with cancelled orders should not be interpreted as revenue lost. A cancelled order's full order value cannot automatically be assumed to represent revenue that the platform would have realized.

Similarly, the compensation figure represents **recorded compensation only**, because compensation information was missing for some cancelled orders.

EDA :

Cancellation Type	Cancelled Orders	Gross Order Value	Share of Platform Gross Order Value	Recorded Restaurant Compensation
Customer Cancelled	89	₹57,062.84	0.39%	₹15,855.82
Zomato Cancelled	81	₹60,289.48	0.41%	₹31,546.65
Total	170	₹117,352.32	0.81%	₹47,402.47

Total platform gross order value: ₹14,554,058.14

Compensation Coverage :

Cancellation Type	Cancelled Orders	Orders with Recorded Compensation	Recorded Compensation
Customer Cancelled	89	58	₹15,855.82
Zomato Cancelled	81	75	₹31,546.65
Total	170	133	₹47,402.47

EDA Interpretation :

A total of **170 orders were cancelled** during the study period, comprising **89 customer-cancelled** and **81 Zomato-cancelled** orders. Together, these cancellations represented **₹117,352.32 in gross order value**, equivalent to **0.81% of the platform's total gross order value of ₹14.55 million**.

The gross order value exposure was relatively similar between the two cancellation types, with **₹57,062.84** associated with customer cancellations and **₹60,289.48** with Zomato cancellations.

Recorded restaurant compensation totalled **₹47,402.47**. Zomato-cancelled orders accounted for **₹31,546.65 (approximately 66.5%)** of this recorded compensation, while customer-cancelled orders accounted for **₹15,855.82**.

However, compensation was recorded for only **133 of the 170 cancelled orders**. Therefore, ₹47,402.47 should be interpreted as **recorded restaurant compensation rather than the total compensation incurred**.

Investigation Conclusion :

Cancelled orders represented a **small share of overall platform gross order value (0.81%)** during the study period. While the absolute exposure was limited relative to total platform order value, Zomato-cancelled orders accounted for a larger share of the **recorded restaurant compensation**, despite having a similar number of cancelled orders to customer cancellations.

This provides useful context for the cancellation analysis: **cancellations appear limited in overall order-value exposure, but platform-initiated cancellations are associated with a larger share of the recorded restaurant compensation.** However, because compensation data is incomplete, this should be treated as an indication of recorded exposure rather than a complete measure of cancellation-related cost.

1.2 Revenue and Customer Value

1.2.1 Does the discount received by customers vary with their overall ratings ?

Why This Matters :

Discounts represent a direct business investment intended to attract customers and improve the value of their orders. Understanding whether customers who receive different discount amounts give different ratings can help assess whether promotional benefits are reflected in customers' overall post-order feedback.

Analytical Note :

This analysis was restricted to **2,491 orders with available customer ratings**. Orders without ratings were excluded because they do not provide the customer feedback required for this analysis.

EDA :

Customer Rating	Number of Rated Orders	Mean Total Discount	Median Total Discount
1	177	94.70	95.18
2	82	127.14	100.00
3	144	87.67	93.00
4	360	115.03	90.00
5	1,728	110.63	90.00

EDA Interpretation :

The average total discount varied somewhat across the customer-rating groups, but there was **no consistent increasing or decreasing pattern**. Median discounts were also relatively similar across the groups.

These descriptive results do not suggest a strong relationship between the discount received and the rating given by the customer. Statistical testing was therefore used to determine whether the observed differences were statistically meaningful.

Statistical Hypotheses :

H₀: The distribution of total discount is the same across the different customer-rating categories.

H₁: The distribution of total discount differs across at least one customer-rating category.

Assumption / Test Selection :

The **Shapiro-Wilk test** was performed separately for each rating group. All groups had **p < 0.05**, indicating that Total Discount was not normally distributed within the rating groups. Histogram inspection also showed right-skewed distributions.

Therefore, the normality assumption required for One-Way ANOVA was not satisfied, so a **Kruskal-Wallis H test** was used to compare total discounts across the five rating groups.

Statistical Validation :

Metric	Result
Test Used	Kruskal-Wallis H Test
H Statistic	4.55
p-value	0.3366
Significance Level (α)	0.05
Decision	Fail to reject H_0

Since $p = 0.3366 > 0.05$, the null hypothesis was not rejected. There is insufficient statistical evidence to conclude that total discount differs across customer-rating categories.

Effect Size :

Epsilon Squared (ϵ^2) = 0.0002

The effect size is **negligible**, indicating that customer rating explains virtually none of the variation in total discount.

Investigation Conclusion :

The analysis found **no statistically significant difference in total discount across customer-rating levels** ($H = 4.55$, $p = 0.3366$). Although discount amounts varied slightly between rating groups, there was no consistent pattern, and the negligible effect size ($\epsilon^2 = 0.0002$) indicates that the practical relationship was extremely weak.

Therefore, **the discount received on an order does not appear to be meaningfully reflected in customers' overall ratings within the available rated orders.**

1.2.2 How is customer spending distributed across the customer base ?

Why This Matters :

Understanding how customer spending is distributed helps identify whether customer value is spread relatively evenly or concentrated among a smaller group of high-value customers. This provides a baseline for understanding the distribution of customer value during the study period.

EDA :

Statistic	Total Spend
Count	11,607
Mean	1,253.90
Standard Deviation	1,518.05
Minimum	53.00
25th Percentile (Q1)	506.63
Median	785.40
75th Percentile (Q3)	1,415.40
Maximum	28,331.05

EDA Interpretation :

Customer spending was **highly right-skewed**, indicating that spending was not evenly distributed across the customer base. Most customers spent relatively modest amounts during the study period, while a smaller group contributed substantially higher spending.

The **median customer spend was ₹785.40**, considerably below the **mean of ₹1,253.90**, indicating that a relatively small number of high-spending customers pulled the average upward. The large gap between the 75th percentile (**₹1,415.40**) and the maximum (**₹28,331.05**) further indicates the presence of customers with substantially higher spending than the majority.

The box plot also showed numerous high-value outliers, reinforcing that customer spending is concentrated among a relatively small group of higher-value customers.

Investigation Conclusion :

Customer spending is **highly concentrated rather than evenly distributed**. Most customers generated relatively modest spending during the study period, while a smaller group contributed substantially higher customer value.

1.2.3 Do a small group of high-spending customers contribute a disproportionately large share of total revenue ?

Why This Matters :

Understanding whether revenue is concentrated among a small group of customers helps the business assess its dependence on high-value customers. This can support decisions around customer retention, loyalty initiatives, and personalized engagement.

EDA :

Customer Segment (by Total Spend)	Share of Total Revenue (%)
Top 10% Customers	37.08
Top 20% Customers	53.50
Top 30% Customers	64.90
Top 40% Customers	73.64
Top 50% Customers	80.57

EDA Interpretation :

Customer revenue was unevenly distributed across the customer base. The **top 10% of customers contributed 37.08% of total revenue**, while the **top 20% contributed 53.50%**.

This means that the remaining 80% of customers contributed **46.50%** of total revenue, while the remaining 90% contributed **62.92%**.

The concentration becomes even clearer as the segment expands: the **top 30% contributed 64.90%**, and the **top 50% contributed 80.57%** of total revenue.

Therefore, revenue appears to be substantially concentrated among higher-spending customers rather than being evenly distributed across the customer base.

Investigation Conclusion :

A relatively small group of high-spending customers contributes a disproportionately large share of platform revenue. In particular, the **top 20% of customers generate more than half (53.50%) of total revenue**, while the top 10% alone generate 37.08%.

This indicates that **high-value customers represent an important part of the platform's revenue base**, making their retention and continued engagement an important business consideration.

1.2.4 Is higher customer spending primarily driven by more frequent ordering or by higher-value orders ?

Why This Matters :

Understanding what drives higher customer spending helps determine whether high-value customers generate more value mainly because they **order more frequently**, place **higher-value orders**, or both. This can help the business decide whether increasing order frequency, average order value, or both should be considered when developing customer-value strategies.

EDA :

Metric	Remaining 80% Customers	Top 20% Customers
Number of Customers	9,286	2,321
Median Orders per Customer	1	3
Median Average Order Value	₹566.02	₹856.44

EDA Interpretation :

Higher customer spending appears to be associated with **both greater order frequency and higher average order value**.

The typical customer in the top 20% spending segment placed **3 orders**, compared with **1 order** among the remaining 80%. At the same time, the median average order value was **₹856.44** for the top 20% compared with **₹566.02** for the remaining customers.

Therefore, high-spending customers do not appear to generate greater value through frequency alone. They also tend to place **higher-value orders**.

Investigation Conclusion :

The analysis suggests that higher customer spending is driven by **two complementary behaviours: customers ordering more frequently and spending more per order**.

This is important for the later Driver Analysis because it means that simply encouraging more orders is **not the only potential route to increasing customer value**. Both repeat purchasing and order value contribute to the difference between high-value and other customers.

1.2.5 Is higher customer spending associated with shorter reorder intervals ?

Why This Matters :

Customers who return sooner have more opportunities to place additional orders over time. Understanding whether high-spending customers tend to reorder more quickly helps determine whether **shorter gaps between purchases** are an important characteristic of higher customer value.

Analytical Note :

Reorder interval was calculated only for customers who placed **at least two orders**, because a customer with a single order does not have a measurable reorder interval.

EDA :

Metric	Remaining 80% Customers	Top 20% Customers
Number of Customers	9,286	2,321
Median Reorder Interval (Days)	23	17
Mean Reorder Interval (Days)	33.06	25.56

EDA Interpretation :

High-spending customers returned to the platform **more quickly** than the remaining customer base.

The median reorder interval was **17 days** for the top 20% spending customers compared with **23 days** for the remaining 80%. The mean showed the same pattern, at **25.56 days versus 33.06 days**.

This suggests that high-value customers are not only ordering more frequently overall but also tend to **return to the platform sooner between purchases**.

Investigation Conclusion :

Higher-spending customers were associated with **shorter reorder intervals**. The six-day difference in median reorder interval and the similar difference in mean reorder interval suggest that faster repeat purchasing is an important behavioural

characteristic of high-value customers.

1.2.6 Do customers with different purchasing frequencies receive different discount levels per order ?

Why This Matters :

Understanding whether discount levels differ across customers with different purchasing frequencies helps evaluate whether the platform's discount allocation varies according to customer purchasing behaviour.

EDA :

Customer Frequency Segment	Number of Orders	Mean Discount	Median Discount	Standard Deviation
One-Time Customer	7,713	₹101.99	₹99.00	₹144.55
Occasional Customer	6,284	₹100.54	₹90.00	₹134.47
Frequent Customer	7,324	₹97.51	₹80.00	₹156.31

EDA Interpretation :

Discount per order showed some variation across customer frequency segments. **One-time customers received the highest median discount (₹99)**, followed by occasional customers (₹90), while frequent customers received the lowest median discount (₹80).

The mean discount showed a similar pattern, although the differences were relatively small compared with the overall variation in discount amounts.

These descriptive results suggest that discount allocation may differ across customer-frequency groups, but the differences need to be statistically evaluated.

Assumption / Test Selection :

- Customer frequency segments are independent groups.
- Discount per order is a continuous variable.
- Discount distribution is right-skewed/non-normal.

Therefore, a **Kruskal-Wallis H test** was used to compare discount distributions across the three customer-frequency segments.

Statistical Hypotheses :

H₀: The distribution of discount per order is the same across one-time, occasional, and frequent customers.

H₁: At least one customer-frequency segment has a different distribution of discount per order.

Statistical Validation :

Metric	Result
Test Used	Kruskal-Wallis H Test
H Statistic	46.442
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H ₀

The test identified a statistically significant difference in discount per order across customer-frequency segments.

Post-hoc Analysis :

Dunn's test with Bonferroni correction found statistically significant differences between **all three pairs**:

Comparison	Adjusted p-value	Significant?
One-Time vs Occasional	0.0028	Yes
One-Time vs Frequent	<0.001	Yes
Occasional vs Frequent	0.0042	Yes

So the statistically significant overall result was supported by differences between each customer-frequency group.

Effect Size :

Epsilon Squared (ϵ^2) = 0.0021

Despite the statistically significant result, the effect size is **negligible**. This means customer purchasing frequency explains very little of the variation in discount amounts.

Investigation Conclusion :

Discount per order differed statistically across the three customer-frequency groups, with **one-time customers receiving the highest median discount (₹99)** and frequent customers the lowest (₹80).

However, the **negligible effect size ($\epsilon^2 = 0.0021$)** shows that these differences are very small relative to the overall variation in discounts.

Therefore, **customer purchasing frequency does not appear to be a practically important determinant of discount allocation**, despite statistically detectable differences between the groups.

1.2.7 Is the platform's revenue primarily driven by one-time customers or repeat customers ?

Why This Matters :

Understanding which customer group contributes the majority of revenue helps identify an important driver of business performance. If repeat customers generate most of the revenue, retaining existing customers and encouraging repeat purchases may be particularly important for long-term revenue growth. If one-time customers contribute a substantial share, customer acquisition remains an important part of the revenue base.

EDA :

Customer Type	Number of Customers	Total Orders	Total Revenue	Revenue Contribution (%)	Average Revenue per Customer
One-Time Customer	7,713	7,713	₹5,390,359.66	37.04%	₹698.87
Repeat Customer	3,894	13,608	₹9,163,698.48	62.96%	₹2,353.29

EDA Interpretation :

One-time customers represented **66.45% of the customer base** and generated 7,713 orders. Despite being the majority of customers, they contributed **₹5.39 million (37.04%)** of total platform revenue.

Repeat customers represented only **3,894 customers**, but generated **13,608 orders** and contributed **₹9.16 million (62.96%)** of total revenue.

The difference is even more pronounced at the customer level: the average revenue per repeat customer was **₹2,353.29**, compared with only **₹698.87** for a one-time customer.

Investigation Conclusion :

Repeat customers were the primary revenue contributors during the study period. Although they represented only about one-third of the customer base, they generated **62.96% of total revenue**.

This provides an important business insight:

A relatively smaller group of repeat customers generates the majority of platform revenue, making repeat customer retention and conversion of one-time customers into repeat customers are important areas for the business.

1.3 Demand & Temporal Patterns

1.3.1 Does customer order demand vary across different times of the day ?

Why This Matters :

Customer order demand can fluctuate throughout the day because of changing customer routines and meal preferences. Identifying peak ordering periods can help food delivery platforms allocate riders, plan restaurant staffing, maintain operational readiness, and design promotions for lower-demand periods.

EDA :

Time of Day	Number of Orders
Evening / Dinner	12,463
Lunch / Afternoon	6,186
Late Night	2,672

EDA Interpretation :

Order demand varied considerably across the different time periods. **Evening/Dinner recorded the highest demand with 12,463 orders**, accounting for more than half of all orders. Lunch/Afternoon followed with **6,186 orders**, while Late Night recorded the lowest demand with **2,672 orders**.

The descriptive results therefore suggest a strong difference in customer ordering behaviour across the day. Statistical testing was used to determine whether the observed distribution differed meaningfully from an equal distribution across the three periods.

Assumption / Test Selection :

Because the analysis involves frequency counts across a single categorical variable—Time of Day—a **Chi-Square Goodness-of-Fit test** was used.

The assumptions were satisfied:

- The data consist of frequency counts.
- The time-of-day categories are mutually exclusive.
- Each order represents an independent observation.
- Expected frequency in each category exceeded 5.

Statistical Hypotheses :

H₀: Customer orders are equally distributed across the different times of the day.

H₁: Customer orders are not equally distributed across the different times of the day.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Goodness-of-Fit Test
χ^2 Statistic	6,923.3435
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H_0

Since $p < 0.001$, the null hypothesis was rejected. There is strong statistical evidence that customer orders are **not equally distributed across the different times of the day**.

Effect Size :

Cohen's $w = 0.5698$

The effect size indicates a **large practical effect**, showing that the differences in order demand across the time periods are substantial rather than merely statistically detectable.

Investigation Conclusion :

Customer order demand varied substantially across the day. **Evening/Dinner was clearly the dominant ordering period**, while Late Night had considerably lower demand. The Chi-Square test confirmed that this distribution was statistically different from an equal distribution, and the **large effect size ($w = 0.5698$)** indicates that the difference is also practically meaningful.

1.3.2 Does customer order demand vary across different days of the week ?

Why This Matters :

Customer order demand may fluctuate across the week because of differences in customer routines and consumption patterns. Understanding whether demand differs by day can help the platform anticipate higher- and lower-volume periods and plan rider allocation, restaurant staffing, and other operational resources accordingly.

EDA :

Day of the Week	Number of Orders
Saturday	3,923
Friday	3,403
Wednesday	3,077
Sunday	2,958
Tuesday	2,885
Thursday	2,879
Monday	2,196

EDA Interpretation :

Customer order demand varied across the days of the week. **Saturday recorded the highest order volume (3,923)**, followed by Friday (3,403), while **Monday recorded the lowest (2,196)**. The remaining days had relatively similar order volumes, ranging from 2,879 to 3,077 orders.

These differences suggest that customer ordering behaviour may vary across the week. Statistical testing was used to determine whether the observed distribution differed meaningfully from an equal distribution across all seven days.

Assumption / Test Selection :

Because the analysis involves frequency counts for a single categorical variable—Day of the Week—a **Chi-Square Goodness-of-Fit test** was used.

The assumptions were satisfied:

- The data consist of frequency counts.
- Each order belongs to one day of the week.
- The categories are mutually exclusive.
- Observations are assumed to be independent.
- Expected frequency in every category exceeded 5.

Statistical Hypotheses :

H₀: Customer order demand is equally distributed across all days of the week.

H₁: Customer order demand varies across at least one day of the week.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Goodness-of-Fit Test
χ^2 Statistic	552.0918
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H ₀

Since **$p < 0.001$** , the null hypothesis was rejected. There is statistical evidence that customer order demand is **not equally distributed across the days of the week**.

Effect Size :

Cohen's $w = 0.1609$

The effect size indicates a **small practical effect**. Therefore, although the differences across days are statistically detectable, their overall magnitude is relatively modest.

Investigation Conclusion :

Customer order demand differed significantly across the days of the week, with **Saturday recording the highest volume and Monday the lowest**. However, the **small effect size ($w = 0.1609$)** indicates that the overall variation across the week is relatively limited.

Therefore, **day of the week represents a secondary demand pattern rather than a major standalone demand driver** during the study period.

1.3.3 Does customer order demand vary across different months ?

Why This Matters :

Understanding how customer order demand changes across months can help the platform anticipate changes in demand, plan resources and staffing, and evaluate whether seasonal patterns require different operational or marketing strategies.

EDA :

Month	Number of Orders
September	4,241
October	4,277
November	4,491
December	4,301
January	4,011

EDA Interpretation :

Customer order demand appeared relatively consistent across the five-month study period. **November recorded the highest order volume (4,491)**, followed by December (4,301), October (4,277), and September (4,241), while **January recorded the lowest (4,011)**.

Although some variation was visible, the differences between months appeared relatively modest. Statistical testing was therefore used to determine whether the observed distribution differed meaningfully from an equal distribution across the five months.

Assumption / Test Selection :

Because the analysis involves frequency counts for a single categorical variable—Month—a **Chi-Square Goodness-of-Fit test** was used.

The assumptions were satisfied:

- The data consist of frequency counts.
- Monthly categories are mutually exclusive.
- Each order represents an independent observation.
- Expected frequency in each category exceeded 5.

Statistical Hypotheses :

H₀: Customer orders are equally distributed across the different months.

H₁: Customer orders are not equally distributed across the different months.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Goodness-of-Fit Test
χ^2 Statistic	27.5796
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H ₀

Since **p < 0.001**, the null hypothesis was rejected. There is statistical evidence that customer order demand was **not equally distributed across the five months**.

Effect Size :

Cohen's w = 0.0360

The effect size indicates a **negligible practical effect**. Therefore, although the monthly differences are statistically detectable, their overall magnitude is extremely small from a practical business perspective.

Investigation Conclusion :

Customer order demand differed statistically across the five months, with **November recording the highest volume and January the lowest**. However, the **negligible effect size ($w = 0.0360$)** indicates that the overall monthly variation was very small.

Therefore, **month does not appear to be a meaningful standalone demand driver during this study period**. The observed November peak and January trough are worth keeping in mind for future monitoring, but the current data do not provide a strong enough pattern to justify major seasonal intervention.

1.3.4 How did customer-initiated and platform-initiated cancellation rates vary over the study period ?

Why This Matters :

Monitoring cancellation rates over time helps determine whether cancellations remained relatively stable or showed notable changes during the study period. Identifying periods with higher customer-initiated or platform-initiated cancellation rates can help the business identify patterns that may require further investigation.

EDA :

Month	Total Orders	Customer-Cancelled Orders	Zomato-Cancelled Orders	Customer Cancellation Rate (%)	Zomato Cancellation Rate (%)	Overall Cancellation Rate (%)
September	4,241	11	33	0.26	0.78	1.04
October	4,277	12	15	0.28	0.35	0.63
November	4,491	26	14	0.58	0.31	0.89
December	4,301	18	13	0.42	0.30	0.72
January	4,011	22	6	0.55	0.15	0.70

EDA Interpretation :

Customer-initiated and platform-initiated cancellations showed different patterns over the study period.

Customer cancellation rates fluctuated between 0.26% and 0.58%, with the highest rate observed in November and the lowest in September. There was no consistent upward or downward trend.

In contrast, **Zomato cancellation rates declined steadily from 0.78% in September to 0.15% in January**. This represents a much clearer temporal pattern than the customer-initiated cancellations.

Despite these variations, the **overall cancellation rate remained below 1.1% in every month**, indicating that cancellations represented a relatively small share of total orders throughout the study period.

These patterns help identify where further investigation may be useful, but descriptive analysis alone cannot establish what caused the changes over time.

Investigation Conclusion :

Customer and platform-initiated cancellations followed **different temporal patterns** during the study period. Customer cancellation rates fluctuated without a clear trend, whereas **Zomato cancellation rates showed a consistent decline from September through January**.

Overall cancellation rates remained relatively low throughout the period. Therefore, the most notable finding here is not a broad increase in cancellation behaviour, but the **declining pattern in platform-initiated cancellations**, which may warrant further investigation into whether operational conditions or processes changed over the study period.

1.3.5 Does discounting vary across different times of the day ?

Why This Matters :

Food delivery platforms may use discounts to influence customer purchasing behaviour and manage demand across different periods. Understanding whether discount levels vary by time of day helps assess whether promotional activity is aligned with demand patterns and whether there are opportunities to optimize discount allocation.

EDA :

The distribution of **Total Discount** was right-skewed, so the **median** was used as the primary measure for comparing discount levels.

Time of Day	Median Total Discount
Late Night	₹97.88
Lunch / Afternoon	₹93.00
Evening / Dinner	₹87.08

EDA Interpretation :

Discount levels showed some variation across the different time periods. **Late Night had the highest median discount (₹97.88)**, followed by Lunch/Afternoon (₹93.00), while **Evening/Dinner had the lowest median discount (₹87.08)**.

This is interesting because Evening/Dinner is the period with the highest customer demand, yet it had the lowest typical discount. Statistical testing was therefore used to determine whether the observed differences were statistically meaningful.

Assumption / Test Selection :

Total Discount was a continuous, right-skewed variable, and the objective was to compare its distribution across three independent time-of-day groups.

Therefore, a **Kruskal–Wallis H test** was used.

The assumptions considered were:

- Total Discount is a continuous variable.
- Observations are independent.
- Time-of-day categories represent independent groups.
- The non-parametric test is appropriate given the non-normal distribution of Total Discount.

Statistical Hypotheses :

H₀: The distribution of Total Discount is the same across different times of the day.

H₁: The distribution of Total Discount differs across at least one time period.

Statistical Validation :

Metric	Result
Test Used	Kruskal–Wallis H Test
H Statistic	166.3292
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H ₀

Since **p < 0.001**, the null hypothesis was rejected. There is statistical evidence that Total Discount differs across at least one time-of-day category.

Post-hoc Analysis :

Dunn's post-hoc analysis showed that:

- **Evening/Dinner differed significantly from Late Night**
- **Evening/Dinner differed significantly from Lunch/Afternoon**
- **Late Night and Lunch/Afternoon did not differ significantly**

Therefore, the statistically detected differences were primarily driven by the lower discount levels observed during Evening/Dinner.

Effect Size :

Epsilon Squared (ϵ^2) = 0.0077

The effect size indicates a **negligible practical effect**. Although the statistical test detected differences between the time periods, their overall magnitude was very small.

Investigation Conclusion :

Discount levels differed statistically across the time periods, with **Evening/Dinner having the lowest median discount** and Late Night the highest. The post-hoc analysis confirmed that Evening/Dinner differed significantly from both other periods.

However, the **negligible effect size ($\epsilon^2 = 0.0077$)** indicates that these differences are very small in practical terms.

Therefore, **time of day does not appear to be a meaningful standalone driver of discount levels**, even though discounting varies statistically across periods.

1.3.6 Does discounting vary across different days of the week ?

Why This Matters :

Food delivery platforms may adjust promotional discounts across different days of the week to influence customer demand and support business objectives. Understanding whether discount levels vary across the week helps evaluate whether promotional activity is aligned with weekly demand patterns and how discounts are distributed across different days.

EDA :

Because Total Discount was right-skewed, the **median** was used as the primary measure for comparing discount levels across the days of the week.

Day	Median Total Discount
Sunday	₹99.00
Monday	₹93.00
Thursday	₹90.00
Wednesday	₹90.00
Saturday	₹89.78
Friday	₹80.00
Tuesday	₹80.00

EDA Interpretation :

Discount levels showed some variation across the week. **Sunday recorded the highest median discount (₹99.00)**, followed by Monday (₹93.00). Thursday and Wednesday had a median discount of ₹90.00, while Saturday recorded ₹89.78. **Friday and Tuesday had the lowest median discount at ₹80.00.**

These differences suggest that promotional discounting may vary across the week. However, the descriptive differences alone do not establish whether the variation is statistically meaningful.

Assumption / Test Selection :

Total Discount was a continuous, right-skewed variable, and the objective was to compare its distribution across **seven independent day-of-week groups**.

Therefore, a **Kruskal–Wallis H test** was used.

The assumptions considered were:

- Total Discount is a continuous variable.
- Observations are independent.
- Day-of-week categories represent independent groups.
- The non-parametric approach is appropriate given the non-normal distribution of Total Discount.

Statistical Hypotheses :

H₀: The distribution of Total Discount is the same across different days of the week.

H₁: The distribution of Total Discount differs across at least one day of the week.

Statistical Validation :

Metric	Result
Test Used	Kruskal–Wallis H Test
H Statistic	208.6370
p-value	< 0.001
Significance Level (α)	0.05
Decision	Reject H ₀

Since **$p < 0.001$** , the null hypothesis was rejected. There is statistical evidence that Total Discount differs across at least one day of the week.

Post-hoc Analysis :

Dunn's post-hoc analysis identified **several significant pairwise differences** between individual days.

This indicates that the statistically significant Kruskal–Wallis result was not simply due to overall variation across all seven groups; specific day-to-day differences were also present.

Effect Size :

Epsilon Squared (ϵ^2) = 0.0095

The effect size indicates a **negligible practical effect**. Therefore, although statistical differences exist between some days, the overall magnitude of variation in discount levels is very small.

Investigation Conclusion :

Total discount per order differed statistically across days of the week, with **Sunday having the highest median discount (₹99.00)** and **Tuesday and Friday the lowest (₹80.00)**. Dunn's post-hoc analysis also identified several statistically significant pairwise differences.

However, the **negligible effect size ($\epsilon^2 = 0.0095$)** indicates that these differences are small in practical terms.

Therefore, **day of the week does not appear to be a practically important determinant of discount levels**, despite statistically detectable differences between some days. This suggests that weekly variation in discounting is relatively

minor compared with the overall variation in discount amounts.

1.4 Customer Retention

1.4.1 How well are newly acquired customers retained over the following weeks ?

Why This Matters :

Retaining customers is important for the long-term growth of a food delivery platform. Understanding how many newly acquired customers continue placing orders after their first purchase helps assess the strength of customer retention and identify whether there is a need to investigate factors associated with repeat behaviour.

Analytical Note :

Customer retention was analyzed for a maximum of **10 weeks after the customer's first order**. Cohorts without a complete 10-week observation period were excluded to ensure fair comparison across acquisition cohorts.

EDA :

Weeks Since First Order	Average Retention (%)
0	100.00
1	8.90
2	8.00
3	6.82
4	6.25
5	5.36
6	5.23
7	5.30
8	5.21
9	4.86
10	4.66

EDA Interpretation :

Customer retention declined substantially after the first order. On average, **8.90% of newly acquired customers returned during the first week after their initial purchase**. Retention then gradually declined, reaching **4.66% by Week 10**.

The largest drop occurred immediately after the first week, while retention remained relatively stable around 5% from approximately Week 5 onward.

This indicates that **only a small proportion of newly acquired customers continue ordering several weeks after their first purchase**, making early repeat behaviour an important area for further investigation.

Investigation Conclusion :

Customer retention showed a clear decline following the first order, falling from **8.90% in Week 1 to 4.66% by Week 10**. The pattern suggests that the platform loses a substantial proportion of newly acquired customers early in their customer journey, while a smaller group continues ordering over subsequent weeks.

Therefore, **early customer retention appears to be an important area for the business to investigate further**, particularly to understand what differentiates customers who return from those who do not.

1.4.2 Do customer retention patterns differ across weekly acquisition cohorts ?

Why This Matters :

Customers acquired at different points in time may show different levels of continued engagement with the platform. Comparing acquisition cohorts helps identify whether retention has remained relatively stable or whether newer customer

cohorts are behaving differently from earlier ones.

If later cohorts show consistently weaker retention, this would indicate that the decline is not simply a general feature of customer behaviour but a pattern associated with when customers entered the platform, which would justify further investigation.

Analysis Approach :

Customer retention was calculated for each weekly acquisition cohort for up to **10 weeks after the customer's first order**.

To make the cohorts comparable, only cohorts with a complete 10-week observation period were included. Retention patterns were then compared across the acquisition cohorts.

Cohort Retention Pattern :

The cohort-level results showed a noticeable difference in retention across acquisition weeks.

Earlier acquisition cohorts generally maintained higher retention across the 10-week period, while later cohorts tended to show lower retention.

For example:

- **Cohort 1** retained **8.31%** of customers by Week 10.
- **Cohort 4** retained **5.98%** by Week 10.
- **Cohort 8** retained **4.32%** by Week 10.
- **Cohort 12** retained only **2.88%** by Week 10.

This suggests that retention performance was not uniform across acquisition cohorts.

EDA Interpretation :

The cohort retention patterns suggest that **later-acquired customers generally exhibited weaker retention during their first ten weeks** than customers acquired earlier in the study period.

The difference is not limited to a single week. Later cohorts tend to remain at lower retention levels across much of the observation period.

However, visual or descriptive differences alone do not establish whether there is a systematic relationship between acquisition cohort and retention. Therefore, a statistical test was used to evaluate whether retention systematically changed across successive cohorts.

Statistical Hypotheses :

H₀ : There is no monotonic relationship between acquisition cohort and average customer retention.

H₁ : There is a monotonic relationship between acquisition cohort and average customer retention.

Statistical Validation :

Test Used: Spearman Rank Correlation

Spearman correlation was selected because acquisition cohort is an **ordered variable**, and the objective was to determine whether retention showed a systematic increasing or decreasing pattern across successive cohorts.

Metric	Result
Spearman's ρ	-0.944
p-value	< 0.001
Decision	Reject H ₀

The analysis found a **strong negative monotonic relationship** between acquisition cohort and average retention ($\rho = -0.944$, $p < 0.001$).

This indicates that **later acquisition cohorts generally had lower retention during their first ten weeks than earlier cohorts**.

Investigation Conclusion :

Customer retention differed substantially across acquisition cohorts. Earlier cohorts generally showed stronger retention, while later cohorts experienced lower retention during their first ten weeks.

The strong negative Spearman correlation ($\rho = -0.944$) provides statistical evidence that retention declined systematically across successive acquisition cohorts during the study period.

However, **this analysis identifies a temporal cohort pattern, not its cause**. The result does not establish that a particular acquisition strategy, promotion, season, customer mix, or operational change caused the decline.

Therefore, the important business finding is:

Retention appears to have weakened across successive customer acquisition cohorts, making the decline in newer-cohort retention an important area for further investigation.

1.4.3 How frequently do customers place orders on the platform ?

Why This Matters :

Understanding how frequently customers place orders helps assess customer engagement and whether the platform primarily relies on one-time customers or repeat customers. This provides a baseline for understanding customer purchasing behaviour and long-term customer value.

EDA :

Number of Orders	Number of Customers	Percentage of Customers (%)
1	7,713	66.45
2	1,906	16.42
3	824	7.10
4	435	3.75
5	254	2.19
6	134	1.15
7	106	0.91
8	59	0.51
9+	166	1.43
Total	11,603	100.00

EDA Interpretation :

The customer order-frequency distribution shows that the platform is **primarily composed of one-time customers**. Approximately **66.45% of customers placed only one order** during the study period.

As order frequency increased, the proportion of customers generally declined. Only **16.42%** placed two orders, while the share continued to fall for customers placing three or more orders.

Customers placing **9 or more orders represented only 1.43%** of the customer base, indicating that highly frequent customers form a relatively small segment.

Investigation Conclusion :

The platform's customer base is heavily weighted toward **one-time and low-frequency customers**, while highly frequent customers represent a small proportion of the overall base.

This establishes an important retention finding:

The main customer-value challenge is not simply maintaining highly frequent customers; it is converting a large pool of one-time customers into repeat customers.

1.4.4 Is order cancellation behaviour different between one-time and repeat customers ?

Why This Matters :

Customer cancellations can affect both customer experience and business performance. Comparing cancellation behaviour between one-time and repeat customers helps determine whether cancellation patterns differ across customer groups and whether either group requires further attention.

EDA :

Customer Type	Total Orders	Customer-Cancelled Orders	Cancellation Rate (%)
One-Time Customer	7,713	27	0.35
Repeat Customer	13,608	62	0.46

EDA Interpretation :

Customer-initiated cancellations were rare in both groups. One-time customers had a cancellation rate of **0.35%**, while repeat customers had a slightly higher rate of **0.46%**.

The difference was only **0.11 percentage points**, so the descriptive results do not suggest a substantial difference in cancellation behaviour between the two customer groups. Statistical testing was therefore used to determine whether the observed difference was statistically meaningful.

Assumption / Test Selection :

Both Customer Type and Customer Cancellation are categorical variables, and the observations are independent. All expected cell frequencies were greater than 5.

Therefore, a **Chi-Square Test of Independence** was used.

Statistical Hypotheses :

H₀: Customer type and customer-initiated order cancellation are independent.

H₁: Customer type and customer-initiated order cancellation are associated.

Statistical Validation :

Metric	Result
Test Used	Chi-Square Test of Independence
χ^2 Statistic	1.078
p-value	0.299
Significance Level (α)	0.05
Decision	Fail to reject H ₀

Since **p = 0.299 > 0.05**, the null hypothesis was not rejected. There is insufficient statistical evidence of an association between customer type and customer-initiated cancellations.

Effect Size :

Cramér's V = 0.0071

The effect size is **negligible**, indicating that customer type explains virtually none of the variation in customer-initiated cancellation behaviour.

Investigation Conclusion :

Customer-initiated cancellation rates were very similar between one-time and repeat customers. Although repeat customers had a slightly higher observed cancellation rate (**0.46% vs. 0.35%**), the difference was **not statistically significant** and the effect size was negligible.

Therefore, **customer type does not appear to be a meaningful standalone factor associated with customer-initiated cancellations during the study period.**

1.5 Restaurant Performance

1.5.1 How does overall business performance vary across different restaurant brands ?

Why This Matters :

Restaurant partners are major supply-side contributors to a food delivery platform. Comparing demand, revenue, repeat engagement, and cancellation metrics across restaurant brands helps identify the strongest contributors, highlight potential operational concerns, and understand how concentrated platform activity is across restaurant partners.

EDA :

Restaurant	Unique Customers	Total Orders	Order Share (%)	Repeat Customer Rate (%)	Total Revenue	Revenue Share (%)	Customer Cancellation Rate (%)
Aura Pizzas	8,442	14,548	68.23	31.52	₹10,751,617.42	73.87	0.37
Swaad	3,268	6,332	29.70	34.12	₹3,545,521.86	24.36	0.49
Tandoori Junction	109	154	0.72	19.27	₹133,665.95	0.92	0.65
Dilli Burger Adda	165	227	1.06	22.42	₹101,709.62	0.70	0.88
The Chicken Junction	31	32	0.15	0.09	₹12,380.99	0.09	0.00
Masala Junction	27	28	0.13	0.06	₹9,162.30	0.06	3.57

EDA Interpretation :

Restaurant performance was **highly concentrated among two brands.**

Aura Pizzas was the dominant restaurant, accounting for **68.23% of platform orders and 73.87% of revenue**. It also served the largest customer base, with 8,442 unique customers.

Swaad was the second-largest contributor, accounting for **29.70% of orders and 24.36% of revenue**. Interestingly, it recorded the highest repeat customer rate among the major restaurants at **34.12%**, compared with 31.52% for Aura Pizzas.

Together, **Aura Pizzas and Swaad accounted for 97.93% of platform orders and 98.23% of revenue**. The remaining restaurants each contributed less than 2% of total orders and revenue.

Cancellation rates for the two major restaurants remained below 1%. The higher rates observed among some smaller restaurants should be interpreted cautiously because their order volumes are very small.

Investigation Conclusion :

The platform's restaurant-side business is **extremely concentrated around Aura Pizzas and Swaad**. These two brands effectively account for almost the entire observed restaurant business during the study period.

This makes restaurant concentration an important business consideration and provides a strong reason to examine **both the performance of these major partners and the platform's dependence on them** in the later Driver Analysis and Recommendation sections.

1.5.2 How do operational efficiency and pricing characteristics differ across restaurant brands ?

Why This Matters :

Restaurant performance is not determined only by order volume and revenue. Differences in kitchen preparation, rider waiting, pricing, packaging charges, and discounting can reveal how restaurant operations and commercial strategies vary across partners.

EDA :

Restaurant	Median KPT (Min)	Median Rider Wait (Min)	Median Packaging Charges	Median Total Discount	Median Order Value
Aura Pizzas	15.98	3.20	₹29.95	₹93.00	₹628.95
Dilli Burger Adda	17.82	2.80	₹17.50	₹150.00	₹367.50
Masala Junction	12.68	2.80	₹13.75	₹0.00	₹288.75
Swaad	17.07	2.80	₹23.00	₹80.00	₹483.00
Tandoori Junction	19.46	3.95	₹40.74	₹0.00	₹719.74
The Chicken Junction	14.73	5.65	₹13.50	₹84.60	₹283.50

EDA Interpretation :

Operational and pricing characteristics varied across restaurant brands.

Among the two dominant restaurants, **Aura Pizzas had a lower median KPT than Swaad (15.98 vs. 17.07 minutes)** but a slightly higher median rider wait time (3.20 vs. 2.80 minutes). Aura also had higher median packaging charges and order value.

The smaller restaurants showed greater variation. **Tandoori Junction had the highest median KPT (19.46 minutes), packaging charge (₹40.74), and order value (₹719.74)**. Meanwhile, **Dilli Burger Adda had the highest median discount (₹150)**.

These differences suggest that restaurant brands follow different operational and pricing patterns. However, the smaller restaurants have very low order volumes, so their metrics should **not be treated as equally reliable comparisons** with Aura Pizzas and Swaad.

Investigation Conclusion :

Restaurant brands differ in both their **operational characteristics and pricing profiles**. However, because almost all platform activity comes from Aura Pizzas and Swaad, the most meaningful restaurant-level investigation should focus primarily on these two major partners rather than treating the small restaurants as equally important.

2. Driver Analysis

2.1 Operational Performance & Cancellation Drivers

Investigation Focus :

The initial analysis identified several operational patterns related to order fulfilment and cancellations. This investigation focused on determining whether **order-ready marking, kitchen preparation time, restaurant, and time of day** could help

explain the most meaningful operational pattern identified: the substantially higher rider wait time associated with problematic order-ready marking.

2.1.1 Order-Ready Marking as an Operational Driver of Rider Wait Time

What We Found :

Orders with problematic order-ready markings had substantially higher rider wait times.

Order Ready Status	Orders	Median Rider Wait
Correctly	19,054	2.50 min
Incorrectly	1,890	7.20 min
Missed	209	3.70 min

Orders marked **Incorrectly** had a median rider wait of **7.20 minutes**, compared with only **2.50 minutes** for correctly marked orders.

The Kruskal–Wallis test also found a statistically significant difference, with $\epsilon^2 = 0.0692$, indicating a meaningful practical difference.

What This Means :

The combination of the large descriptive difference and statistical evidence suggests that **order-ready marking is an important operational factor associated with rider waiting**.

However, this does **not** establish that incorrect marking itself causes the additional waiting. The next step was therefore to investigate what conditions might be associated with problematic marking.

Is Problematic Order-Ready Marking Concentrated in Particular Restaurants?

Problematic marking was examined across restaurants by combining **Incorrect** and **Missed** statuses.

Restaurant	Incorrect + Missed
Tandoori Junction	14.94%
Masala Junction	14.28%
Dilli Burger Adda	14.09%
The Chicken Junction	12.50%
Swaad	10.75%
Aura Pizzas	10.24%

Finding :

Problematic marking occurred across **multiple restaurant partners** rather than being concentrated in one major restaurant.

Among the two dominant partners, the rates were relatively close:

- **Swaad:** 10.75%
- **Aura Pizzas:** 10.24%

The apparently higher rates among smaller restaurants should be treated cautiously because some are based on extremely small order volumes.

Interpretation :

The data therefore **does not provide strong evidence for a single restaurant-specific problem**. Restaurant-level differences are useful as monitoring signals, but they are not strong enough to attribute the underlying issue to a particular restaurant.

Could Kitchen Preparation Time Explain Problematic Marking ?

Finding :

Median KPT increased slightly across the marking categories:

- Correctly: **16.30 min**
- Incorrectly: **16.88 min**
- Missed: **17.50 min**

The statistical test produced $p = 0.018$, but the effect size was only $\epsilon^2 = 0.0003$.

Interpretation :

Although the difference was statistically detectable, it was **extremely small in practical terms**. The median difference between correctly and incorrectly marked orders was only **0.58 minutes**.

Therefore:

KPT does not appear to be a meaningful standalone driver of problematic order-ready marking in this dataset.

This is an important distinction : statistical significance does not automatically make a factor operationally important.

Could Time of Day Explain Problematic Marking ?

The marking rates were:

Time of Day	Correctly	Incorrectly	Missed
Evening/Dinner	90.07%	8.49%	1.44%
Late Night	89.33%	8.61%	2.06%
Lunch/Afternoon	88.49%	9.81%	1.70%

The Chi-square test found a statistically significant association ($\chi^2 = 15.5752$, $p = 0.0036$), but **Cramér's V = 0.0191**, indicating a negligible practical effect.

Interpretation :

Time of day therefore does **not appear to be a meaningful standalone driver** of problematic order-ready marking.

Importantly, the busiest period, **Evening/Dinner**, did not have the highest problematic-marking rate. This weakens the idea that higher order demand alone explains the marking problem.

Overall Driver Assessment :

The investigation can be summarized as:

Factor investigated	Evidence	Practical importance	Assessment
Order-ready marking → Rider wait	Strong statistical evidence	Meaningful effect	Important operational relationship
Restaurant → Problematic marking	Differences observed	Small restaurants have unstable rates	No clear restaurant-specific driver
KPT → Problematic marking	Significant p-value	$\epsilon^2 = 0.0003$	Negligible
Time of day → Problematic marking	Significant p-value	$V = 0.0191$	Negligible

Overall Finding :

The investigation found a clear and practically meaningful relationship between **problematic order-ready marking and rider wait time**. However, when we investigated potential explanations for why problematic marking occurs, the available variables did not provide strong evidence for a specific underlying driver.

KPT showed a statistically significant difference, but its effect was negligible. Time of day also showed a statistically significant association, but again with a negligible effect. Restaurant-level differences were observed, but problematic

marking occurred across multiple restaurants and the smaller restaurants had insufficient volume to support reliable ranking.

Business Takeaway :

Problematic order-ready marking should be treated as an operational process-monitoring opportunity because of its meaningful relationship with rider waiting. However, the available data does not justify attributing the problem to a specific restaurant, preparation-time level, or time period.

2.1.2 Customer Cancellation Drivers

Investigation Focus :

Customer cancellations can lead to lost order value, operational disruption, and potential customer dissatisfaction. The initial analysis showed that cancellations were relatively rare, so the investigation focused on identifying whether any operational, temporal, geographic, or customer-related factor showed a meaningful relationship with customer cancellation.

The main factors investigated were **Kitchen Preparation Time, Time of Day, Delivery Distance, Acquisition Cohort, and Operational Progression**.

What factors were associated with customer cancellations?

What We Found -

The investigation did **not identify any factor as a strong standalone driver** of customer cancellation.

Factor	Main Finding	Evidence	Assessment
Kitchen Preparation Time	Cancelled orders actually had slightly lower KPT	p = 0.3105; small effect	Deprioritize
Time of Day	Late Night had the highest cancellation rate (0.67%)	p = 0.0112; negligible effect	Monitor
Delivery Distance	Long-distance orders had the highest cancellation rate (0.84%)	Significant association; negligible effect	Monitor
Acquisition Cohort	November had the highest 30-day cancellation rate (0.72%)	p = 0.3693; negligible effect	Deprioritize
Operational Progression	Most cancellations occurred before substantial fulfilment activity	p = 0.08; small effect	Descriptive signal

These findings are based on the detailed investigations documented in the analysis.

What the individual investigations tell us :

Kitchen Preparation Time -

Customer-cancelled orders did **not** have longer preparation times. Their median KPT was actually slightly lower than non-cancelled orders, and the difference was not statistically significant.

Conclusion: KPT does not appear to be a meaningful standalone driver of customer cancellation.

Time of Day -

Late Night had the highest customer cancellation rate at **0.67%**, compared with **0.52% during Lunch/Afternoon** and **0.31% during Evening/Dinner**.

The association was statistically significant, but the effect size was negligible.

Conclusion: Time of day is interesting enough to **monitor**, but the evidence is too weak to treat it as a major cancellation driver.

Delivery Distance -

Orders of **9 km or more** had the highest observed cancellation rate at **0.84%**, compared with roughly **0.32%–0.45%** for the other distance groups.

However, the effect size was negligible.

Conclusion: Long-distance orders are worth keeping as a **monitoring signal**, but distance does not appear to be a strong standalone driver of cancellation.

Acquisition Cohort -

First-30-day cancellation rates varied between **0.40% and 0.72%** across acquisition cohorts, but there was no consistent pattern. The statistical test also found no significant association.

Conclusion: There is no meaningful evidence that the month in which customers were acquired influenced their early cancellation behaviour.

Operational Progression -

Most customer cancellations occurred relatively early in the order process. **75.4%** were classified as either having no recorded operational activity or compensation only.

However, the difference in restaurant compensation across operational patterns was not statistically significant.

Conclusion: This gives useful context about **when cancellations happen**, but it does not establish operational progression as a driver of customer cancellation.

Overall Finding :

No factor investigated showed strong evidence of being a standalone driver of customer cancellation. Time of day and delivery distance showed some interesting patterns, particularly higher cancellation rates during Late Night and among longer-distance orders, but their practical effects were very weak. Kitchen Preparation Time and acquisition cohort did not show meaningful evidence of influencing cancellation behaviour. Operational progression provided additional context, showing that most customer cancellations occurred before substantial fulfilment activity had been recorded.

Business Takeaway :

The analysis does **not support targeting one specific operational factor as the primary cause of customer cancellations**.

Instead:

- **Late Night** → retain as a monitoring signal
- **Long-distance orders** → retain as a monitoring signal
- **KPT** → no reason to prioritize based on this dataset
- **Acquisition cohort** → no reason to prioritize
- **Operational progression** → useful for understanding cancellation exposure, but not a validated driver

This means the business should **not immediately redesign operations around any one of these factors**. The available evidence is better suited to identifying areas for continued monitoring and future investigation.

Important Limitation :

Only **89 customer-cancelled orders** were available in the dataset. Therefore, a non-significant or weak result should be interpreted as **insufficient evidence of a meaningful relationship in this sample**, rather than proof that a factor can never influence customer cancellation.

2.1.3 Zomato Cancellation Drivers

Investigation Focus :

Zomato-initiated cancellations can create lost order-value exposure, restaurant compensation costs, and operational disruption. The investigation therefore examined whether **Kitchen Preparation Time, Rider Wait Time, Order Value, Time of Day, and Restaurant Subzone** showed meaningful relationships with Zomato cancellations.

The aim was not simply to find statistically significant differences, but to determine whether any factor had enough **practical importance** to be considered an actionable driver.

What factors were associated with Zomato cancellations?

Factor	Main Finding	Statistical / Practical Evidence	Assessment
Kitchen Preparation Time	Zomato-cancelled orders had slightly lower KPT	p = 0.009; small effect	Low priority
Rider Wait Time	Wait times were almost identical	p = 0.8373; negligible effect	Deprioritize
Total Order Amount	Higher mean but lower median for cancelled orders	p = 0.2108; negligible effect	Deprioritize
Time of Day	Late Night had highest cancellation rate (0.898%)	p < 0.001; negligible effect	Monitor
Restaurant Subzone	Sector 4 had highest observed rate (0.521%)	p = 0.0518; negligible effect	Deprioritize

Operational Factors :

Kitchen Preparation Time -

Zomato-cancelled orders actually had **lower** KPT than non-cancelled orders:

- Median KPT: **14.72 min** vs. **16.35 min**

The difference was statistically significant, but the effect was small.

Finding -

Although KPT showed a statistically detectable difference, the direction of the difference does not support the idea that longer kitchen preparation is driving Zomato cancellations. Zomato-cancelled orders actually had slightly lower preparation times.

Driver Status: Low priority.

Rider Wait Time -

Rider wait times were almost identical :

- Zomato-cancelled: **3.50 min median**
- Non-cancelled: **3.10 min median**

The statistical test found **no significant difference (p = 0.8373)** and the effect size was negligible.

Finding -

The data provides no meaningful evidence that rider wait time is associated with Zomato cancellations.

Driver Status: Deprioritize.

Order Value -

Zomato-cancelled orders had:

- Higher mean order value: **₹744.31** vs. **₹682.38**
- Lower median order value: **₹544.95** vs. **₹597.45**

Because order value was highly right-skewed, the median provides a better picture of the typical order.

The statistical test was not significant (**p = 0.2108**) and the effect size was negligible.

Finding -

The higher mean among cancelled orders is largely influenced by greater variability rather than a meaningful difference in typical order value. The evidence does not support order value as an important standalone driver of Zomato cancellations.

Driver Status: Deprioritize.

Time of Day -

Zomato cancellation rates varied noticeably across time periods:

Time of Day	Zomato Cancellation Rate
Late Night	0.898%
Evening / Dinner	0.321%
Lunch / Afternoon	0.275%

Late Night had the highest observed cancellation rate.

The association was statistically significant, but the effect size was **negligible (Cramér's V = 0.0321)**.

Finding -

Late Night shows an interesting cancellation pattern, but the very small effect size means that time of day does not appear to be a strong standalone driver of Zomato cancellations.

Driver Status: Monitor.

Restaurant Subzone -

Cancellation rates varied across subzones, with **Sector 4 recording the highest observed rate at 0.521%**.

However, the statistical test did not find a significant association (**p = 0.0518**) and the effect size was negligible.

Some subzones also had extremely small order volumes, so their observed rates should not be interpreted as reliable evidence of better or worse performance.

Finding -

The available evidence does not support restaurant subzone as a meaningful standalone driver of Zomato cancellations.

Driver Status: Deprioritize.

Overall Zomato Cancellation Driver Summary -

What We Found :

The investigation did **not identify a strong standalone factor associated with Zomato cancellations**.

KPT and Time of Day produced statistically significant results, but their practical effects were small or negligible. Rider wait time, order value, and restaurant subzone did not show meaningful evidence of association.

Overall Finding :

Zomato cancellations do not appear to be strongly explained by any single operational, temporal, geographic, or order-value factor examined in this analysis. Late Night showed the clearest pattern worth keeping an eye on, but its practical effect was still very small.

Business Takeaway :

The available evidence does not justify an immediate intervention targeting any one of the investigated factors. **Late Night cancellation behaviour can be retained as a monitoring signal**, while KPT, rider wait time, order value, and restaurant subzone do not currently warrant prioritization as standalone drivers.

Important Limitation :

Only **81 Zomato-cancelled orders** were observed compared with more than 21,000 non-cancelled orders. Therefore, these findings should be interpreted as **evidence from this dataset**, not proof that these factors can never influence Zomato

cancellations.

2.2 Demand & Temporal Drivers

Purpose :

Customer demand was not evenly distributed throughout the day, week, or study period. This section examines the strongest temporal demand patterns and then looks at related **pricing, restaurant, and operational patterns** to understand the business context around those differences.

2.2.1 Time of Day → Customer Demand

Business Finding :

Evening/Dinner was clearly the dominant demand period, accounting for **58.46% of all orders (12,463 orders)**.

Lunch/Afternoon accounted for 29.03%, while Late Night accounted for only 12.53%.

The Chi-square goodness-of-fit test confirmed that order demand was not equally distributed across the three periods ($\chi^2 = 6923.34$, $p < 0.001$). The large effect size (**Cohen's $w = 0.5698$**) shows that this is not just a statistically detectable difference—the variation is also meaningful.

Key Pattern :

| **Evening/Dinner is the clear peak-demand period and should be treated as the primary temporal demand driver.**

Driver Status :

| **Time of Day → Customer Demand: Priority Driver**

What Happens Around the Evening Demand Peak?

The strong evening demand pattern raised the next question:

| **Is the evening peak associated with differences in discounts, restaurant demand, spending, or operational performance?**

The purpose here was **not to claim why evening demand exists**, but to understand what other observable patterns accompany it.

A. Promotional Discount

Discount levels differed statistically across time periods ($H = 166.33$, $p < 0.001$), but the effect size was negligible ($\epsilon^2 = 0.0077$).

Interestingly, **Evening/Dinner had the lowest median discount (₹87.08)**, while Late Night had the highest (₹97.88).

Finding -

| The evening demand peak does **not** appear to be explained by higher per-order discounts. In fact, the highest-demand period had the lowest median discount.

Status: Monitor / Contextual Finding

B. Restaurant-Level Demand Pattern

The next question was whether the evening peak was mainly being created by one or two restaurants.

Evening/Dinner represented the largest share of orders for most restaurants:

- Aura Pizzas — **59.97%**
- Swaad — **55.31%**
- Tandoori Junction — **57.79%**
- Dilli Burger Adda — **51.10%**

- Masala Junction — **64.29%**

The Chicken Junction was the exception, where Evening/Dinner and Lunch/Afternoon each accounted for 43.75%.

Finding :

The evening demand peak is **not concentrated in a single restaurant**. It appears across most restaurant brands, including the two major restaurant partners.

For the major restaurants, Evening/Dinner was also generally associated with higher median order values. However, the smaller restaurants had very low order volumes, so their patterns should be treated cautiously.

Business Interpretation :

The evidence suggests that the evening demand peak is a **platform-wide temporal pattern rather than a restaurant-specific phenomenon**.

C. Operational Context

Since Evening/Dinner is the busiest period, the next question was whether this higher demand creates noticeable operational pressure.

Kitchen Preparation Time -

KPT differed statistically across time periods, but the practical effect was negligible ($\epsilon^2 = 0.0071$).

Evening/Dinner had a median KPT of **16.33 minutes**, compared with 16.82 minutes during Lunch/Afternoon and 15.32 minutes during Late Night.

Rider Wait Time -

Rider wait time also differed statistically, but the practical effect was negligible ($\epsilon^2 = 0.0124$).

Interestingly, **Lunch/Afternoon had the highest median rider wait (4.3 minutes)**, while Evening/Dinner was only 2.6 minutes.

Zomato Cancellation -

Late Night had the highest Zomato cancellation rate (**0.898%**), compared with 0.321% during Evening/Dinner and 0.275% during Lunch/Afternoon.

Although the association was statistically significant, the effect was negligible (**Cramér's V = 0.0321**).

Operational Finding :

The busiest period does not appear to experience substantially worse operational performance. Evening/Dinner has the highest demand, but it does not have the highest KPT, rider wait time, or Zomato cancellation rate.

Overall Demand & Temporal Driver Interpretation -

The analysis established a **strong and practically important evening demand peak**. Evening/Dinner accounts for nearly **58.5% of all orders**, making it the most important temporal demand period in the dataset.

Further investigation showed that:

- The evening peak is visible across **most restaurant brands**.
- Major restaurants generally have **higher typical order values** during Evening/Dinner.
- Evening/Dinner does **not** have unusually high discounts; it actually has the lowest median discount.
- Higher evening demand is **not accompanied by substantial operational deterioration**.
- KPT, rider wait time, and Zomato cancellation show some statistical differences across periods, but their practical effects are negligible.

Overall Business Finding -

Evening/Dinner is the strongest temporal demand driver in the dataset. The peak appears to be broad-based across restaurants rather than being driven by a single restaurant, and it is not accompanied by substantially higher discounts or operational deterioration.

What We Can and Cannot Conclude :

The analysis tells us **when demand is highest and what patterns accompany that demand**, but it does not establish **why** customers order more during Evening/Dinner. Factors such as customer routines, meal preferences, promotions, or other external factors were not directly measured in the dataset.

Therefore, the evening demand peak should be treated as a **strong planning signal**, not as evidence that any particular factor causes the higher demand.

Business Implication :

The strong concentration of orders during Evening/Dinner makes this period important for **rider availability, restaurant capacity, staffing, and operational readiness**. At the same time, the relatively stable operational metrics suggest that the platform and restaurants were able to handle the higher evening demand without a major deterioration in the measured operational indicators.

2.2.2 Day of Week → Customer Order Demand

Business Question :

Does customer order demand vary across different days of the week?

What We Found :

Customer demand varied across the week, with **Saturday recording the highest order volume (3,923)** and **Monday the lowest (2,196)**. Most of the remaining days were relatively close to each other.

The overall difference was statistically significant, but the practical effect was small.

Statistical Evidence :

The Chi-square goodness-of-fit test found a statistically significant difference in order distribution across the days of the week ($\chi^2 = 552.09$, $p < 0.001$). However, **Cohen's w = 0.1609**, indicating a **small practical effect**.

Where Does the Difference Come From ?

To understand the significant result more closely, we examined how much each day contributed to the overall Chi-square statistic.

Day	Observed Orders	Expected Orders	χ^2 Contribution
Saturday	3,923	3,045.86	252.60
Monday	2,196	3,045.86	237.13
Friday	3,403	3,045.86	41.88
Thursday	2,879	3,045.86	9.14
Tuesday	2,885	3,045.86	8.50
Sunday	2,958	3,045.86	2.53
Wednesday	3,077	3,045.86	0.32

The difference was driven mainly by **Saturday and Monday**, which together contributed approximately **89% of the total Chi-square statistic**. Saturday had substantially more orders than expected, while Monday had substantially fewer.

Business Interpretation :

The main weekly demand pattern is the contrast between a Saturday peak and a Monday trough. However, the overall effect is much smaller than the strong time-of-day pattern identified earlier. Day of week is therefore useful for routine demand planning, but it does not appear to be a major standalone demand driver.

Driver Status: Secondary Driver / Monitor

2.2.3 Month → Customer Order Demand

Business Question :

Does customer order demand vary across different months?

What We Found :

Monthly demand remained **relatively stable** during the five-month study period. November recorded the highest order volume (**4,491**), while January recorded the lowest (**4,011**).

Although November and January differed from the other months, the overall differences were small.

Statistical Evidence :

The Chi-square goodness-of-fit test found a statistically significant difference in the monthly order distribution ($\chi^2 = 27.58$, $p < 0.001$). However, **Cohen's $w = 0.0360$** , indicating a **negligible practical effect**.

Where Does the Difference Come From ?

To understand the statistically significant result, we examined the contribution of each month to the Chi-square statistic.

Month	Observed Orders	Expected Orders	χ^2 Contribution
January	4,011	4,264.2	15.03
November	4,491	4,264.2	12.06
December	4,301	4,264.2	0.32
September	4,241	4,264.2	0.13
October	4,277	4,264.2	0.04

The contribution analysis shows that **January's lower demand and November's higher demand accounted for almost all of the observed difference**, contributing approximately **98.7% of the total Chi-square statistic**. The other three months were very close to the expected distribution.

Business Interpretation :

There is some month-to-month variation, particularly around the November peak and January trough, but the overall effect is negligible. Therefore, month does not appear to be a meaningful standalone demand driver within this five-month study period. These patterns can be monitored as more data becomes available, but they do not currently justify major operational changes.

Driver Status: Low Priority / Monitor

2.3 Customer Retention Drivers

Investigation Focus :

Customer retention declines as customers move further away from their first order. The analysis also showed that **later acquisition cohorts generally had lower retention**.

The next step was therefore to investigate whether differences in **customer experience, first-order experience, or acquisition conditions** could help explain this pattern.

Retention After the First Order :

What We Found -

Retention declined quickly after the first order.

Week Since First Order	Average Retention
Week 1	8.90%
Week 2	8.00%
Week 3	6.82%
Week 4	6.25%
Week 5	5.36%
Week 6	5.23%
Week 7	5.30%
Week 8	5.21%
Week 9	4.86%
Week 10	4.66%

Only 8.90% of newly acquired customers returned in the first week, and retention gradually fell to 4.66% by week 10. The largest decline occurred during the earlier weeks, after which retention remained at a much lower level.

Business Interpretation :

The platform is losing a large proportion of newly acquired customers early in their lifecycle. This makes the first few weeks after acquisition an important period for retention monitoring and customer engagement.

Retention Across Acquisition Cohorts :

The next question was whether this retention pattern was similar for customers acquired at different points in the study period.

What We Found -

Later acquisition cohorts generally showed lower retention over their first 10 weeks.

Spearman rank correlation showed a strong negative relationship between acquisition cohort and average retention:

- **Spearman's $\rho = -0.944$**
- **$p < 0.001$**

Customers acquired in later cohorts generally showed weaker retention than customers acquired earlier in the study period.

Business Interpretation -

Retention was not only declining with customer age; the retention performance of newer acquisition cohorts was also weaker. This suggests that something about the conditions surrounding later customer acquisition may be worth investigating.

Important Limitation -

This analysis identifies a strong cohort pattern, but it does **not establish why later cohorts had lower retention**. The cohort effect could reflect differences in customers, acquisition channels, promotions, seasonality, or other factors that

are not fully captured in the dataset.

Investigating Why Retention Declined :

Since retention was weaker for later cohorts, we investigated three possible explanations:

1. **Does customer experience deteriorate as customers place more orders ?**
2. **Was the first-order experience different for customers who returned?**
3. **Did acquisition conditions change across cohorts?**

2.3.1 Customer Experience Across Order Progression

Question :

Does customer experience change as customers progress through later orders?

We compared **KPT, rider wait time, and discount** across different order stages.

Although statistical tests detected differences in all three variables, the effect sizes were very small.

Finding -

Customer experience did not show a meaningful deterioration as customers progressed through later orders.

Therefore, the changes in KPT, rider wait time, and discount across order stages **do not provide a strong explanation for the decline in retention.**

Driver Status : Not supported as a meaningful explanation

2.3.2 First-Order Experience of Returning vs. Non-Returning Customers

The previous analysis looked at what happened during later orders. We therefore looked at the customer's **very first delivered order** to see whether returning customers started with a different experience.

Customers were divided into:

- **Returned:** placed at least one additional delivered order within 28 days
- **Did not return:** placed no additional delivered order within 28 days

We compared:

- KPT
- Rider wait time
- Total discount
- Total order value

What We Found :

- **KPT:** no significant difference
- **Rider wait time:** no significant difference
- **Discount:** no significant difference
- **Order value:** statistically significant, but effect size was very small (**rank-biserial = -0.049**)

Finding :

Customers who returned and those who did not had broadly similar first-order experiences across the factors examined. The small difference in order value was statistically detectable but too small to have meaningful practical importance.

Business Interpretation

The first-order operational and promotional experience examined here does not provide a strong explanation for why some customers return while others do not.

Driver Status: Not supported as a meaningful explanation

2.3.3 Acquisition Conditions Across Cohorts

Since later cohorts had weaker retention, we then checked whether customers joining the platform at different times were experiencing substantially different conditions during their first order.

We compared:

- KPT
- Rider wait time
- Total discount
- Total order value
- Time of day
- Day of week

What We Found :

KPT, rider wait time, and order value did not show a consistent pattern alongside the decline in cohort retention. Discount levels were lower in some later cohorts, particularly December and January, but the pattern was not consistent across all cohorts. Time-of-day distributions remained broadly similar, while day-of-week composition showed some variation without a clear relationship with the retention decline.

Business Interpretation :

The acquisition conditions examined in the dataset do not show a strong or consistent pattern that explains why later cohorts had weaker retention.

Driver Status: No clear explanation identified

2.3.4 Overall Retention Finding

The investigation produced **two clear findings**:

1. Retention is weak and declines over time

Only **8.90%** of customers returned in the first week, falling to **4.66% by week 10**.

2. Later acquisition cohorts performed worse

Later cohorts generally showed lower retention, with a strong negative relationship between cohort timing and average retention ($\rho = -0.944$, $p < 0.001$).

2.3.5 What We Investigated

We then tested whether this pattern could be explained by:

Customer experience

→ No meaningful deterioration found

First-order experience

→ Returning and non-returning customers were broadly similar

Acquisition conditions

→ No strong or consistent pattern found across cohorts

2.3.6 Overall Driver Conclusion

The analysis confirms that customer retention is a significant business concern: customers are most likely to drop off during the early weeks after acquisition, and later acquisition cohorts show weaker retention. However, the factors available in this dataset do not provide a strong explanation for why this is happening.

KPT, rider wait time, discounts, order value, and the other acquisition conditions examined did not show a strong enough relationship with retention to be considered meaningful standalone drivers.

Business Takeaway :

The priority is therefore not to assume that operational delays or discounts are causing the retention problem. The stronger conclusion is that retention itself requires attention, while the underlying reason remains unresolved in the current dataset.

Further investigation would require additional customer-level information such as acquisition source, customer preferences, restaurant choices, customer feedback, or other behavioural information that is not available in the current dataset.

Driver Status :

Customer Retention → Confirmed Business Concern / Root Cause Not Identified

2.4 Customer Value Drivers

Investigation Focus :

Customer spending was not evenly distributed across the customer base. The analysis therefore focused on understanding **where customer value comes from** and what characteristics distinguish higher-value customers from the rest of the customer base.

The investigation examined:

- overall spending concentration
- one-time vs. repeat customers
- order frequency
- average order value
- reorder behaviour

2.4.1 Customer Spending Is Concentrated

What We Found :

Customer spending was highly right-skewed.

The **median customer spent ₹785.40**, compared with a mean of **₹1,253.90**. The large difference between the two, together with the presence of high-spending outliers, shows that customer value was not evenly distributed.

Most customers generated relatively modest spending, while a smaller group of customers contributed substantially more.

This raised the next question:

Is this concentration large enough to matter to overall platform revenue?

2.4.2 Repeat Customers Are More Important to Revenue

The customer base was divided into one-time and repeat customers.

Customer Type	Customers	Orders	Revenue	Revenue Contribution
One-Time	7,713	7,713	₹5.39M	37.04%

Customer Type	Customers	Orders	Revenue	Revenue Contribution
Repeat	3,894	13,608	₹9.16M	62.96%

Finding :

Repeat customers represented only 33.55% of customers but generated 62.96% of total revenue. In contrast, one-time customers represented 66.45% of customers but contributed only 37.04% of revenue.

Business Interpretation :

Revenue is disproportionately dependent on customers who return and place multiple orders. This makes repeat usage an important characteristic of customer value.

This naturally leads to the next question:

What makes these repeat and high-value customers more valuable?

2.4.3 High-Spending Customers Contribute Disproportionately

The customer base was ranked according to total spending.

Customer Segment	Share of Total Revenue
Top 10%	37.08%
Top 20%	53.50%
Top 30%	64.90%
Top 40%	73.64%
Top 50%	80.57%

Finding -

The **top 20% of customers generated 53.50% of total revenue**, while the top 10% alone generated 37.08%.

This confirms that the concentration observed in the spending distribution is not simply caused by a handful of extreme outliers.

Business Interpretation -

A relatively small group of high-value customers contributes a disproportionately large share of revenue. Understanding what distinguishes these customers is therefore important for customer-value strategy.

2.4.4 What Makes High-Value Customers Different?

The top 20% of customers by total spending were compared with the remaining 80%.

Metric	Remaining 80%	Top 20%
Median Orders per Customer	1	3
Median Average Order Value	₹565.96	₹855.75

Both order frequency and average order value showed statistically significant differences:

Metric	p-value	Rank-Biserial Correlation
Order Frequency	< 0.001	0.803
Average Order Value	< 0.001	0.488

What We Found :

High-spending customers were different in **two ways**: they ordered more frequently and they spent more per order.

However, the difference in **order frequency was substantially stronger** than the difference in average order value.

Business Interpretation :

Ordering more frequently appears to be the stronger characteristic of high customer value, although higher order values also contribute.

This leads to a further question:

Do high-value customers order more frequently because they return to the platform sooner?

2.4.5 Do High-Value Customers Return Sooner?

Reorder intervals were compared between customers in the top 20% by spending and the remaining 80%.

Only customers with at least two orders were included because a single-order customer has no measurable reorder interval.

Metric	Remaining 80%	Top 20%
Median Reorder Interval	23 days	17 days
Mean Reorder Interval	33.06 days	25.56 days

The difference was statistically significant ($p < 0.001$), but the effect size was small (**rank-biserial correlation = 0.073**).

Finding :

High-spending customers tended to return sooner than the remaining customer base. However, the small effect size indicates that reorder interval is not a strong differentiator of high customer value by itself.

Business Interpretation :

This supports the earlier finding that high-value customers are more engaged with the platform and tend to purchase more frequently. However, **shorter reorder intervals alone do not explain why a customer becomes high value.**

2.4.6 Overall Customer Value Finding

The analysis gives us a clear picture of where customer value comes from:

1. Revenue is concentrated

The **top 20% of customers generate 53.50% of revenue.**

2. Repeat customers matter disproportionately

Only **33.55% of customers are repeat customers**, but they generate **62.96% of revenue.**

3. Frequency is the strongest differentiator

High-value customers typically place **more orders**, and the difference in order frequency is stronger than the difference in average order value.

4. Order value also matters

High-value customers also have a higher typical order value (**₹855.75 vs. ₹565.96**).

5. High-value customers tend to return sooner

Their median reorder interval is shorter, but the effect is relatively small.

2.4.7 Business Interpretation

The strongest characteristic associated with customer value is repeat purchasing. High-value customers are not simply customers who spend a lot on individual orders; they tend to return more often and also have somewhat higher order values.

Therefore, the findings support a business focus on **building sustained repeat usage and retaining valuable customers**, rather than relying only on increasing individual order value.

Important Limitation :

These analyses identify characteristics associated with higher customer value, but they do **not establish causation**. For example, the analysis does not prove that reducing reorder intervals will cause customers to become more valuable, or that increasing order frequency will automatically increase revenue.

2.5 Restaurant Performance Analysis

Restaurant partners are an important part of the platform's business. After looking at demand, cancellations, customer retention, and customer value, we next looked at the restaurant side of the business.

The aim was to understand **which restaurants contribute most to the platform, how their operations and pricing differ, and whether customer behaviour changes across restaurants**.

2.5.1 Overall Restaurant Performance

Business Question:

How does overall business performance vary across different restaurant brands?

Approach:

We compared restaurants based on customer reach, order volume, revenue contribution, repeat customer rate, and cancellation rates.

Key Finding:

Restaurant performance was highly concentrated among **Aura Pizzas and Swaad**.

- Aura Pizzas accounted for **68.23% of all orders** and **73.87% of total revenue**.
- Swaad accounted for another **29.70% of orders** and **24.36% of revenue**.
- Together, the two restaurants contributed **97.93% of platform orders** and **98.23% of revenue**.
- Swaad had the highest repeat customer rate at **34.12%**, slightly above Aura Pizzas at **31.52%**.

The remaining restaurants contributed very little to overall platform activity. Their higher or lower cancellation and repeat-customer rates therefore need to be treated cautiously because their order volumes were much smaller.

Business Interpretation:

The platform's restaurant business is overwhelmingly dependent on Aura Pizzas and Swaad. Their performance is therefore much more important to overall platform performance than that of the smaller restaurant partners.

This raised the next question: **apart from their size, how different are these restaurants in the way they operate and price their orders?**

2.5.2 Operational and Pricing Differences Across Restaurants

Business Question:

How do operational efficiency and pricing characteristics differ across restaurant brands?

We compared **kitchen preparation time, rider wait time, packaging charges, discounts, and order value** across restaurants.

Key Finding:

Restaurant brands showed clear differences in their operational and commercial characteristics.

Among the two major restaurants:

- Aura Pizzas had a lower median KPT than Swaad (**15.98 vs. 17.07 minutes**).
- Aura Pizzas had slightly higher median rider wait time (**3.20 vs. 2.80 minutes**).
- Aura Pizzas also had higher median packaging charges (**₹29.95 vs. ₹23.00**) and order value (**₹628.95 vs. ₹483.00**).

The smaller restaurants showed even greater variation. For example:

- Tandoori Junction had the highest median KPT (**19.46 minutes**) and highest median order value (**₹719.74**).
- Dilli Burger Adda had the highest median discount (**₹150**).
- Masala Junction had a median discount of **₹0**.

However, the smaller restaurants had very few orders, so these differences should not be treated as reliable restaurant rankings.

Business Interpretation:

Restaurants do not follow the same operational or pricing pattern. Their preparation times, rider waits, discounts, packaging charges, and order values differ. However, the two major restaurants are the most important for platform-level decisions because they account for almost all business activity.

This led to the next question: **do customers also behave differently across restaurants depending on when they place their orders?**

2.5.3 Demand and Customer Behaviour Across Time of Day

Business Question:

How do customer demand and order value vary across different times of the day for each restaurant?

We compared the **share of orders and median order value** across Lunch/Afternoon, Evening/Dinner, and Late Night for each restaurant. We also looked at discounts to see whether higher demand during a particular period was accompanied by higher promotional activity.

Key Finding:

The **Evening/Dinner period was the main demand period for most restaurants.**

For the two major restaurants:

- **59.97%** of Aura Pizzas orders were placed during Evening/Dinner.
- **55.31%** of Swaad orders were placed during Evening/Dinner.

This shows that the strong evening demand seen at the overall platform level was also visible across the major restaurant partners.

Evening/Dinner also generally had the highest median order value for most restaurants. For example:

- Aura Pizzas: **₹670.53**
- Swaad: **₹528.67**

However, the smaller restaurants showed some different patterns, and their low order volumes make those differences less reliable.

Discounts did not show one consistent pattern across restaurants and time periods. This suggests that the evening demand peak was **not simply explained by restaurants offering higher discounts during that period.**

Business Interpretation:

Evening/Dinner is the main demand period across most restaurants and is generally associated with higher order values. However, discounting does not follow one common peak-period pattern across restaurants.

This means that peak demand appears to be a broader customer-demand pattern rather than something that can simply be explained by higher discounts.

2.5.4 Overall Restaurant Finding

The restaurant analysis gives us three main findings:

1. Restaurant performance is highly concentrated.

Aura Pizzas and Swaad together account for almost all platform orders and revenue.

2. Restaurants operate and price differently.

KPT, rider wait time, discounts, packaging charges, and order values vary across restaurant brands.

3. Evening/Dinner is the dominant demand period.

This pattern is visible across most restaurants, and evening orders generally have higher order values. However, discounts do not show a consistent increase during this period.

Business Interpretation:

The restaurant side of the platform is driven mainly by a small number of major partners, but these restaurants do not behave exactly the same way. Their operational and pricing characteristics differ, while customer demand is generally concentrated in the evening.

For the platform, this suggests two priorities: maintaining strong performance among the major restaurant partners and ensuring sufficient restaurant capacity and operational readiness during peak Evening/Dinner demand.

At the same time, the analysis does not support a single strategy for all restaurants. Smaller partners show different patterns, but their low order volumes mean that stronger conclusions would require more data.